



**KTH Industrial Engineering  
and Management**

# ESTIMATING INVESTMENT NEEDS FOR THE POWER SECTOR IN THE AFRICAN REGION

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Uppskattning av investeringsbehov i Afrikanska elsektorer

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### Sammanfattning

Detta examensarbete bygger på ett samarbete mellan KTHs avdelning Energisystemanalys, KTH-dESA, och Afrikanska Utvecklingsbanken, AfDB. Arbetet har utförts både från KTH i Stockholm samt på AfDBs tillfälliga huvudkontor i Tunis, Tunisien. Samarbetet mellan KTH-dESA och AfDB är relaterat till tidigare projekt. AfDB har samarbetat med Världsbanken vilket bland annat resulterat i AICD (*African Infrastructure Country Diagnostic*) som i sin tur skapat programmet AIKP (*African Infrastructure Knowledge Program*). Liksom i AICD är en viktig del av AIKP att uppskatta infrastrukturinvesteringar kopplade till elsektorn.

Syftet med detta examensarbete är att skapa matematiska modeller över elsektorerna i afrikanska länder. Resultaten från dessa ska visa vilka investeringar, eltekniker och bränslen som ger det lägsta totala nuvärdet för respektive land för modelleringsperioden (2010-2030 med fokus på perioden 2014-2020), givet de använda parametrarna. Dessa modeller ska sedan användas för att leverera data till en internetapplikation som AfDB nyligen övertagit från Världsbanken. För att skapa modeller används mjukvaran ANSWER OSeMOSYS, vilket är ett transparent program som är utvecklat för att användas vid systemoptimering som sträcker sig över längre tidsperioder (Howells et al., 2011). Data till modellerna skulle tillhandahållas av AfDB, men då denna inte blev tillgänglig i tid har tillfällig data, delvis tillhandahållen av KTH-dESA, använts. Även antalet undersökta länder har reducerats till fyra, Mauritius, Madagaskar, Egypten och Zimbabwe, istället för de ursprungliga 18 länderna.

Modellerna för de fyra länderna blev funktionella, även om resultaten från simuleringarna ännu inte är helt tillförlitliga. Detta då fokus legat på att skapa modeller snarare än resultat, eftersom modellerna i ett senare skede kommer fyllas med ny uppdaterad data från AfDB. I samtliga länder är vatten- och kolkraftverk vanligt förekommande tekniker i resultaten. Egyptens resultat skiljer sig från övriga länders i och med landets stora användning av naturgas i elkonversion. Även Mauritius resultat särskiljer sig genom en relativt stor andel el från decentraliserad PV.

Utöver dessa modeller har även en mall för resultatredovisning skapats i form av två mindre rapporter för Mauritius respektive Egypten. Kontakter har även gjorts med olika personer på AfDB för att underlätta för det fortsatta samarbetet mellan KTH-dESA och AfDB.

Bilden på försättsbladet är hämtad från <http://www.eps-egypt.com/> 2014-08-04 (bilden är beskuren)





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Master of Science Thesis

MJ210x 2014

Estimating Investment Needs for the Power Sector  
in the African Region

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### Abstract

This master thesis is based on collaboration between the division of Energy System Analysis at the Swedish Royal Institute of Technology, KTH-dESA, and the African Development Bank, AfDB. The work has been carried out both at KTH (Stockholm, Sweden) and at AfDB (Tunis, Tunisia). The KTH-dESA – AfDB collaboration is related to previous projects, for example AIKP, the *African Infrastructure Knowledge Program*, which in turn is the result of a previous collaboration with the World Bank that led to the comprehensive knowledge program AICD, *African Infrastructure Country Diagnostic*. As in AIKP, a key part of AICD is to estimate infrastructure investment needs for the power sector.

The main objective of this thesis is to create mathematical models of the power sectors in African countries. The output of these models shows, given the parameters used, what investments, power technologies and fuels that gives the lowest net present value for the modelling period (2010-2030 with focus on 2014-2020). The models developed will then be used to populate an Internet application hosted by AfDB. The application was earlier hosted by the World Bank. To develop the models, the free and open source software ANSWER OSeMOSYS is used. The program is developed in order to make long-run optimization models (Howells et al., 2011). Input data to the models was initially supposed to be provided by AfDB. This data was not available in time why temporary data, partly provided from KTH-dESA, is used and the number of countries investigated in this thesis is reduced to four instead of the original 18 countries.

The models of the four countries are functional, even though the results of the simulations are not yet final. This is caused by the fact that focus has been put on the creation of the models rather than the results, since the models are to be re-populated with up-to-date data from AfDB. Generally, hydropower and coal STPP are commonly used power sources in the simulation results. The results of Egypt have a large share of technologies fueled by natural gas, which distinguishes the country from the other three. Also the Mauritius result differs, due to a relatively large share of power from off-grid PVs.

The remaining objectives of the project, i.e. the completion of two minor country reports and a input data table for AfDB as well as establishing contacts at AfDB, are completed successfully but are not presented in this report.

Front page picture from <http://www.eps-egypt.com/> 2014-08-04 (cropped)

# FOREWORD

This paper is the result of a master thesis in Sustainable Energy Technology that was conducted during spring 2014. The work was carried out at the division of Energy System Analysis (KTH-dESA) and is based on cooperation between the division and the African Development Bank (AfDB). During the process that resulted in this paper, several people have contributed with their time and knowledge. We want to extend a special thanks to a few people whose help have been essential in our work.

*Prof. Mark Howells, KTH-dESA, for being enthusiastic*

*Mr. Oliver Broad, KTH-dESA, for his help in a comprehensive number of questions*

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*Mr. Nirina Letsara, African Development Bank, for the warm reception and your help*

*Mr. Vignesh Sridharan, KTH-dESA, for data and software skills*

We would also want to thank the rest of the team at KTH-dESA and the employers at AfDB who have helped us during our project. Finally, we will thank family members and friends that helped us with practical issues before and during our stay at the African Development Bank.

Stockholm, August 2014

Sandra Davidsson & Anna-Klara Hagberg



# ABBREVIATIONS

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Below, the abbreviations and units used in this paper are listed in alphabetic order.

|          |                                                                                          |
|----------|------------------------------------------------------------------------------------------|
| AfDB     | <i>African Development Bank</i>                                                          |
| AICD     | <i>African Infrastructure Country Diagnostic</i>                                         |
| AIKP     | <i>African Infrastructure Knowledge</i>                                                  |
| CCGT     | <i>Combined Cycle Gas Turbine</i>                                                        |
| CEB      | <i>Central Electricity Board (Mauritius)</i>                                             |
| CSP      | <i>Concentrated Solar Power</i>                                                          |
| EAC      | <i>East Africa Community</i>                                                             |
| EAPP     | <i>East African Power Pool</i>                                                           |
| GW       | <i>Giga (<math>10^9</math>) watt</i>                                                     |
| HFO      | <i>Heavy Fuel Oil</i>                                                                    |
| IPP      | <i>Independent power producer</i>                                                        |
| KTH-dESA | <i>The division of Energy System Analysis at the Royal Institute of Technology (KTH)</i> |
| mUSD     | <i>Million (<math>10^6</math>) US\$, 2010</i>                                            |
| NG       | <i>Natural Gas</i>                                                                       |
| NPV      | <i>Net Present Value</i>                                                                 |
| OCGT     | <i>Open/Single Cycle Gas Turbine</i>                                                     |
| PJ       | <i>Peta (<math>10^{15}</math>) joule</i>                                                 |
| PV       | <i>Photovoltaic cell</i>                                                                 |
| RES      | <i>Reference Energy System</i>                                                           |
| STPP     | <i>Steam Turbine Power Plant</i>                                                         |



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# 1 INTRODUCTION

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This project aims to create functional models for the power sectors of some African countries. In this paragraph more information about the project is presented as well as objectives, delimitations and general information about the countries investigated.

## 1.1 Background

Today, less than 25 % of the Sub-Saharan rural and urban households have access to electricity (World Bank, 2013a). It is stated that there is a clear link between poverty reduction and access to energy services (UN-Energy, 2005 & World Bank, 2005). Electricity plays a vital role in education, gender equality, maternal health conditions and communication (UN-Energy, 2005). Thus it can be stated that large investments in the African power sector is needed. However, all kinds of large scale investment need to be carefully investigated in order to secure the relevance and adequacy of the objects invested in. Energy system modelling offers a valuable possibility to estimate what energy sources which will be most profitable in the future, thereby helping decision makers to start new projects.

This master thesis is done at the KTH division of Energy System Analysis (KTH-dESA). The division has an ongoing collaboration with African Development Bank (AfDB), which as the name reveals is a development bank with the objective to “*spur a sustainable economic development and social progress in its regional member countries*” (AfDB Group, 2014a). 2010 AfDB developed a program called Africa Infrastructure Knowledge Program (AIKP), which aims to “*ensure that critical data and analysis of Africa’s infrastructure sectors will continue into the future on a sustainable basis*” (AfDB Group, 2011a). The program builds on and continues the Africa Infrastructure Country Diagnostic (AICD) which is the result of a previous cooperation between AfDB and the World Bank. AICD covers main infrastructure parameters such as air transport, information and communication technologies, irrigation, ports, power, railways, roads, sanitation and water. This also includes the African power sector. AIKP is, as mentioned, the successor project to AICD but with a long term sustainable basis and a goal to institutionalize the collected data (AfDB Group, 2011a). As a part of AICD, a web-application has been created where the user can change parameters in a chosen power sector and see what investments “should” be executed in the country of interest (AfDB Group, 2011b). This application was earlier hosted by the World Bank but is now taken over by AfDB. To populate the application with results, AfDB has an ongoing cooperation with KTH-dESA.

OSeMOSYS (Open Source energy Modelling SYStem) is a free and open source modelling software for long-run optimization and energy planning (Howells et al., 2011). The software plays a vital part in this project. More information about the software is to be found in paragraph 2.1 *Modelling framework and structure*.

## 1.2 Objectives

The main overall objective of this project is to develop mathematical programming models of the power systems of 18 African countries. The output of those models are then to be used in a web application hosted by AfDB, where the user can change different parameters and see the outcome in form of power investment needs for the country/countries investigated. The interim objectives of this project have changed during the process for different reasons, including misunderstandings between KTH-dESA and AfDB about what data and information was available. The original objectives are listed below.

- Develop new models for Comoros, Madagascar, Mauritius, Sao Tome and Principe, Seychelles, South Sudan and Sudan. Update existing models for Algeria, Burundi, Djibouti, Egypt, Libya, Mali, Morocco, Togo, Tunisia, Swaziland and Zimbabwe
  - Create a functional OSeMOSYS model for each country that can be run with different data setup.
  - Simulate the years 2010-2030 (focus on 2014-2020) and present the results, i.e. find the electricity mix and investments with the lowest total net present value for the period.
- Create a template for country specific result reports to be used for evaluation of preliminary results.
- Assist in the development of AfDBs web-application
  - Together with the web team of AfDB and KTH-dESA the web application should be populated with the results of the OSeMOSYS models.

During the project, the aims changed slightly even though the main objective remained unaffected. The major changes are listed below.

- Due to lack of data, the number of models to be populated, initially by non-AfDB data, and simulated was reduced to four; Egypt, Mauritius, Madagascar and Zimbabwe.
- Create a parameter table together with AfDB for AfDB to send to authorities in the countries to be investigated. This to gather the data needed for the OSeMOSYS models.
- Facilitate future work between KTH-dESA and AfDB by establishing links, for example concerning discussions about the web application and the extension of the AfDB – KTH-dESA contract.
- The work with populating the web-application is delayed (no longer a part of this master thesis) since the in-house knowledge about it is still very low (2014, August). This is caused by the AfDBs recent takeover of the application which earlier was hosted by the World Bank.

### **1.3 Limitations**

This project alone will not provide all the results needed by AfDB. The models built only consider a baseline scenario. To provide all data needed for the web application, each model needs to be run several times, stepwise, changing the parameters that are of in the web application. This quantitative model running will be executed by KTH-dESA.

As mentioned in the earlier sub-paragraph, the number of modelled countries is limited to four. The remaining models will be developed by KTH-dESA when AfDB can provide the data needed.

For the existing power plants, the models do mainly consider grid connected plants since off-grid plants usually are less documented. The models are based on the assumption that there is only one grid per country.

It should be clear that the models created in this project are just models and not a perfect representation of reality. Models are a great tool in order to make reality easier to understand, explain and predict but can never be held as 100 % true. Obvious examples in this project are the limited number of fuels and technologies investigated, as well as the projections of electricity demand. The results of this project can still be used as an indicator of what energy investments that seems to have a potential of being feasible.

## 1.4 Country information

This paragraph presents some characteristics of Egypt, Madagascar, Mauritius and Zimbabwe, including some basic information about each power sector. All the countries that are included in the final project can be seen in Figure 1. Detailed data about these four countries is to be found in *Appendix B-Default data* and *Appendix C- Country specific data* and basic facts about the remaining countries is to be found in *Appendix A- Basic country information*.



Figure 1: Map over Africa and parts of the Middle East with the countries investigated marked in red. Edited from (Google Maps, 2014)

### 1.4.1 EGYPT

|                         |                                                                                                |                                                                                                                                     |
|-------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Official name:          | <b>ARAB REPUBLIC OF EGYPT</b>                                                                  |                                                                                                                                     |
| Region:                 | Northern Africa*                                                                               |                                                                                                                                     |
| Government type:        | Republic                                                                                       |                                                                                                                                     |
| Currency:               | Egyptian Pound (EGP)                                                                           |                                                                                                                                     |
| Surface area, land:     | 1 001 450 km <sup>2</sup>                                                                      |                                                                                                                                     |
| Population (2014 est.): | 86 895 099                                                                                     |                                                                                                                                     |
| GDP/capita (2013 est.): | 551.4 GUSD**                                                                                   |                                                                                                                                     |
| Exports                 |                                                                                                |                                                                                                                                     |
| - Total (2013 est.):    | 24.81 billion USD                                                                              |                                                                                                                                     |
| - Commodities:          | Crude oil and petroleum products, cotton, textiles, metal products, chemicals, processed food. |                                                                                                                                     |
| Imports                 |                                                                                                |                                                                                                                                     |
| - Total (2013 est.):    | 59.22 billion USD                                                                              |                                                                                                                                     |
| - Commodities:          | Machinery and equipment, foodstuffs, chemicals, wood products, fuels.                          | Unspecified references (including map): (CIA, 2014)<br>*Geographical region by (UN Statistics Division, 2013)<br>**Data in 2013 USD |



Egypt is located in the northern Africa and is bordered by Libya to the west, Sudan to the south and Israel and Gaza to the east. The landscape is dominated by deserts and fertile slopes by the Nile. Egypt has two seasons; a mild winter from November to April and a hot summer which occurs between May and October (CIA, 2014). Egypt is considered to be a lower middle income country (World Bank, 2014a) with an economy resting on crude oil and petroleum products, cotton, textiles, metal products, chemicals and processed food (CIA, 2014). Egypt is well electrified, with 99 % of the households are connected to an electricity system (AfDB Group, 2010). The total primary energy supply in Egypt 2009 was 3015.1 PJ. Of this, natural gas represents 50 %, petroleum products 45 %, coal 1 %, biomass 2 % and hydro 2 %. The renewable energy resources in the country are high for wind and solar power (IRENA, 2014). The electricity is mainly converted from fossil fuels as seen in Figure 2.

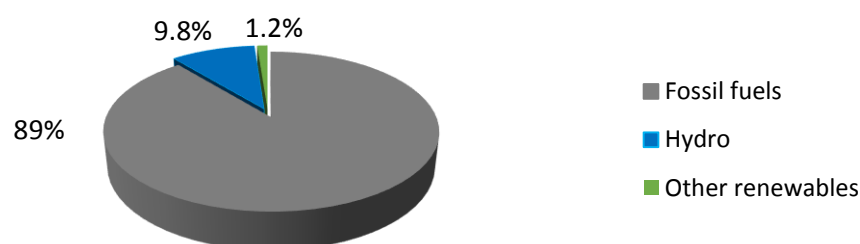


Figure 2: Electricity mix for Egypt 2011, (AfDB Group, 2013).

Egypt's electricity market is regulated by the Ministry of Electricity and Energy (MOEE). Under this agency is a company called Electricity Holding Company (EEHC) which consist of sixteen subsidiaries, divided into six power generation companies, nine electricity distribution companies and one Transmission Company. Egypt is also a member in East African Power Pool (EAPP) (AfDB Group, 2010).

#### 1.4.2 MADAGASCAR

|                                                            |                                                                                         |                        |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------|
| Official name:                                             |                                                                                         | REPUBLIC OF MADAGASCAR |
| Region:                                                    | Eastern Africa*                                                                         |                        |
| Government type:                                           | Republic                                                                                |                        |
| Currency:                                                  | Madagasy Ariary (MGA)                                                                   |                        |
| Surface area, land:                                        | 581 549 km <sup>2</sup>                                                                 |                        |
| Population (2014 est.):                                    | 23 201 926                                                                              |                        |
| GDP/capita (2013 est.):                                    | 1 000 USD**                                                                             |                        |
| Exports                                                    |                                                                                         |                        |
| - Total (2013 est.):                                       | 644.4 million USD                                                                       |                        |
| - Commodities:                                             | Coffee, vanilla, shellfish, sugar, cotton cloth, clothing, chromite, petroleum products |                        |
| Imports                                                    |                                                                                         |                        |
| - Total (2013 est.):                                       | 2.794 billion USD                                                                       |                        |
| - Commodities:                                             | Capital goods, petroleum, consumer goods, food                                          |                        |
| Unspecified references (including map & flag): (CIA, 2014) |                                                                                         |                        |
| *Geographical region by (UN Statistics Division, 2013)     |                                                                                         |                        |
| **2013 USD                                                 |                                                                                         |                        |



Madagascar is a relatively large country located in the Indian Ocean east of Mozambique. The climate varies over the island and is tropical along the coasts, temperate inland and arid in the south of the country (CIA, 2014). The country is considered as a low income country (World Bank, 2014a) and is suffering from deforestation and erosion caused by the use of wood as a primary source of energy.

Year 2008 the total primary energy supply was 274.2 PJ, of which biomass represented 90 %, petroleum products 9 %, hydro 1 % and coal 0.1 %. The country has high renewable energy resources of wind, solar, hydro and ocean power (IRENA, 2014). The electricity is mainly converted from hydropower, which can be seen in Figure 3.

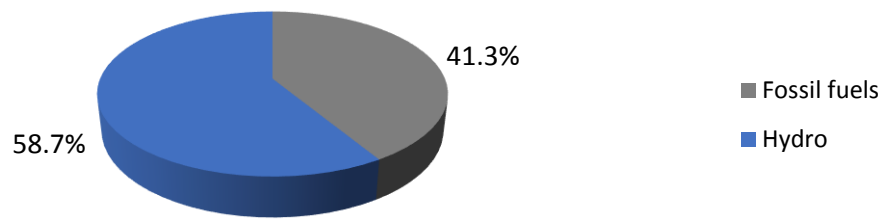


Figure 3: Electricity mix of Madagascar 2011, data from (AfDB Group, 2013).

The electricity in Madagascar is mainly supplied by the state-managed company Jiro sy Rano Malagasy (JIRAMA) and they have three independent grids (Antananarivo Grid, Toamasina Grid and Fianarantsoa Grid). The grid is connected to hydro, HFO and diesel power plants. Beyond this there are also 13 more places that have mini-grids/off-grid power plants (Engineering and consulting firms association, 2009).

#### 1.4.3 MAURITIUS

|                         |                                                                                  |                                                                                                                                                                                                                              |  |
|-------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Official name:          |                                                                                  | <b>REPUBLIC OF MAURITIUS</b>                                                                                                                                                                                                 |  |
| Region:                 | Eastern Africa*                                                                  |                                                                                                                                           |  |
| Government type:        | Parliamentary democracy                                                          |                                                                                                                                                                                                                              |  |
| Currency:               | Mauritian (MUR)                                                                  |                                                                                                                                                                                                                              |  |
|                         | Rupee                                                                            |                                                                                                                                                                                                                              |  |
| Surface area, land:     | 2 030 km <sup>2</sup> **                                                         |                                                                                                                                                                                                                              |  |
| Population (2014 est.): | 1 331 155                                                                        |                                                                                                                                                                                                                              |  |
| GDP/capita (2013 est.): | 16 000 USD***                                                                    |                                                                                                                                                                                                                              |  |
| Exports                 |                                                                                  |                                                                                                                                                                                                                              |  |
| - Total (2013 est.):    | 2.788 billion USD                                                                |                                                                                                                                                                                                                              |  |
| - Commodities:          | Clothing & textiles, sugar, cut flowers, molasses, fish, primates (for research) |                                                                                                                                                                                                                              |  |
| Imports                 |                                                                                  |                                                                                                                                                                                                                              |  |
| - Total (2013 est.):    | 4.953 billion USD                                                                | Unspecified references (including map): (CIA, 2014)<br><br>*Geographical region by (UN Statistics Division, 2013)<br>**Includes Agalega Islands, Cagados Carajos Shoals (Saint Brandon) and Rodrigues<br>***Data in 2013 USD |  |
| - Commodities:          | Manufactured goods, capital equipment, foodstuffs, petroleum products, chemicals |                                                                                                                                                                                                                              |  |

Mauritius is an island nation located east of Madagascar in the Indian Ocean. The country has been colonized by several countries but regained its independence from the United Kingdom in 1968 (CIA, 2014). Mauritius is considered to be an upper middle income country (World Bank, 2014a) with an economy resting on sugar, tourism, textiles and financial services. Bagasse, a rest product from sugar production, also contributes to the county's power production. The climate is tropical with a dry and warm winter between May and November and a hot and humid/rainy summer November to May (CIA, 2014). The total primary energy supply in Mauritius 2010 was 59.0 PJ. Of this, petroleum products represent 54 %, coal 29 %, biomass 16 % and hydro 0.6 %. The renewable energy resources in the country are high for wind and solar power (IRENA, 2014). The electricity is mainly converted from fossil fuels, as can be seen in Figure 4.

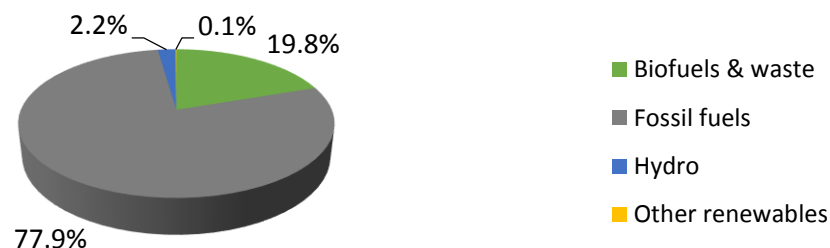


Figure 4: Electricity mix for Mauritius 2011, data from (AfDB Group, 2013).

Central Electrical Board (CEB) is a Government-owned company which is responsible for generation, transmission, distribution and sale of electricity in Mauritius. They have prioritized the expansion of the national grid and have relied on hydro and thermal power generation with imported fossil fuels for its electricity production. Because of the expensive fossil fuel prices, collaboration was commenced between the Sugar Plantations and CEB. In 2008, CEB generated 37% of the electricity while IPPs (Independent Power Producers) produced the other 63 % (Central Electricity Board, 2013).

#### 1.4.4 ZIMBABWE

|                         |                                                                                           |                                                                                                                                    |
|-------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Official name:          | <b>REPUBLIC OF ZIMBABWE</b>                                                               |                                                                                                                                    |
| Region:                 | Eastern Africa*                                                                           |                                                  |
| Government type:        | Parliamentary democracy                                                                   |                                                                                                                                    |
| Currency:               | Zimbabwean dollars (ZWD)                                                                  |                                                                                                                                    |
| Surface area, land:     | 390 847 km <sup>2</sup>                                                                   |                                                                                                                                    |
| Population (2014 est.): | 13 711 721                                                                                |                                                                                                                                    |
| GDP/capita (2013 est.): | 10.48 billion USD**                                                                       |                                                                                                                                    |
| Exports                 |                                                                                           |                                                                                                                                    |
| - Total (2013 est.):    | 3.144 billion USD                                                                         |                                                                                                                                    |
| - Commodities:          | Platinum, cotton, tobacco, gold, ferroalloys and textiles/clothing                        |                                                                                                                                    |
| Imports                 |                                                                                           |                                                                                                                                    |
| - Total (2013 est.):    | 4.571 billion USD                                                                         | Unspecified references (including map & flag): (CIA, 2014)<br>*Geographical region by (UN Statistics Division, 2013)<br>**2013 USD |
| - Commodities:          | Machinery and transport equipment, other manufactures, chemicals, fuels and food products |                                                                                                                                    |

Zimbabwe is located in south-eastern part of Africa and consists mostly of savannah. The climate is sub-tropical with a rainy summer season during November and March (CIA, 2014). Even though Zimbabwe is a country with mineral and coal resources it is considered to be a low income country (World Bank, 2014a).

Year 2009 the total primary energy supply was 398.3 PJ, of which biomass represented 69 %, coal and coal products 20 %, oil and oil products 7 % and hydro 4 %. The country has high renewable energy resources of solar and hydro power (IRENA, 2014). The electricity is generated from fossil fuels and hydro as can be seen in Figure 5.

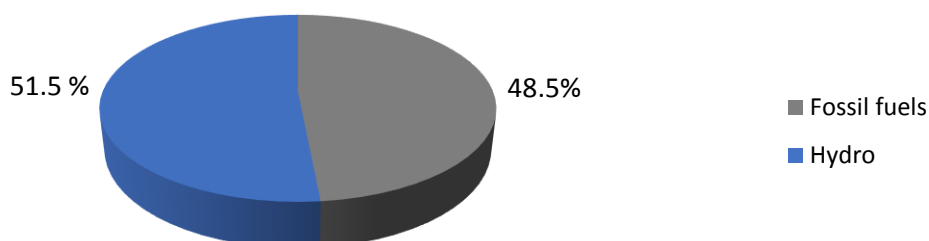


Figure 5: Electricity mix of Zimbabwe 2011, data from (AfDB Group, 2013).

Zimbabwe Power Company (ZPC) is responsible for all generating stations and for the supply of power to the transmission lines. Zimbabwe Electricity Transmission Company (ZETCO) is responsible for the transmission and Zimbabwe Electricity Distribution Company (ZEDC) is responsible for the distribution in the country (AfDB Group, 2012).

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## 2 METHODOLOGY

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In this paragraph, the approach used in the project is described. An explanation of the software used is presented, followed by information about the model structure and assumptions.

### 2.1 Modelling framework and structure

In this project the software OSeMOSYS is used. The paragraph below gives information about the software, followed by paragraphs presenting scenarios and the general model structure as well as the parameters used in the models.

#### 2.1.1 ANSWER OSeMOSYS

To model the energy system the software OSeMOSYS (Open Source energy Modelling SYStem) is used. OSeMOSYS is a free and open source modelling system for long-run optimization and energy planning. The software has several advantages compared to other established energy system model programs, for example the software is free and potentially requires a shorter learning and operating period. The code of the program is relatively transparent and straightforward for experienced energy researchers. OSeMOSYS is built using linear programming method and the programming language GNU Mathprog, which is an open source and freely available programming language. (Howells et al. 2011) In this project the software is used in the interface ANSWER.

In order to make an OSeMOSYS model easy to adapt to a specific analysis, the model is developed in series of blocks. Each block represents functionality and consists of four levels of abstraction; a plain description in English, an algebraic translation of the description in English, programming language describing the model's implementation and at last the application of the model. In the version of OSeMOSYS used in this project, there are in total seven blocks, specifying the model objective, costs, storage, capacity adequacy, energy balance, constraints and emissions. Those seven blocks are further explained below (Howells et al. 2011).

**1 Objective**

*The objective of the model is to minimize the total undiscounted cost of the given energy system and at the same time meet the demand stated by the user.*

**2 Costs**

*To meet the mentioned objective, the cost of each technology is calculated. This is done for each year and each region of the model. The cost is a sum of operating costs (fixed and variable), capital costs and salvage costs.*

**3 Storage**

*This block deals with questions related to storage but is not applicable on the modelling done in this project.*

**4 Capacity adequacy**

*The objective of this block is to establish the total capacity available for each time slice and year in order to ensure that there is enough capacity to meet the demand.*

**5 Energy balance**

*The "production", use and demand of energy need to be feasible annually and at each time slice. The energy balance is also separated in "A" and "B", where energy balance "A" accounts for balancing of time slices and energy balance "B" accounts for annual balancing.*

**6 Constraints**

*In order to limit for example the total capacity, new investments, annual activity or the model period activity for a particular technology, maximum and minimum limit can be used. These constraints belong to this block.*

## 7 Emissions accounting

*It is not unusual that technologies modelled in OSeMOSYS have impacts on the environment “in real world”. This block makes it possible to reflect emissions to the model if the analyst chooses to specify the emissions per unit of activity for each mode of operation of a technology. If this functionality is applied, emission penalties and limits are also specified in this block.*

### 2.1.2 Scenario description

The model will be adjusted in order to produce data for a web application hosted by AfDB. In the previous version of the application it considered factors such as GDP growth rate, price of coal and oil, urban and rural target access rates, a capex cost multiplier and climate change/precipitation sensitivity (AfDB Group, 2011b). Parameters linked to the considered factors need to be changed stepwise in the models, providing new results for all possible combinations/scenarios. These factors are not yet fully defined in terms of parameters affected, ranges etc.

Further discussions between AfDB and KTH-dESA are needed in order to decide how the remaining scenarios should be designed. In this report, the results of a Baseline scenario are described. This scenario assumes that the predicted electricity demands will always be met and that the most feasible mix of technologies/investments/fuels will be used. The scenario takes into account predicted price and demand changes as well as projects that are planned or under construction. However, it does not consider policies linked to for example pollution or carbon dioxide emissions.

### 2.1.3 General model structure

Below in Figure 6 a simplified Reference Energy System, RES, shows the general model structure. In the RES, vertical lines symbolize energy carriers (fuels) whereas the energy converting technologies are symbolized by boxes. All models are driven by the energy demand of the end-users. The demand can be met either by electricity generated from centralized power plants or decentralized off-grid power plants. The different power plants run on fuels that are either imported or domestically extracted/refined. The real RES for a country is more complex than the simplified version below, with several fuels, power plants etc. A more detailed RES can be found in *Appendix E – Detailed Reference Energy System*.

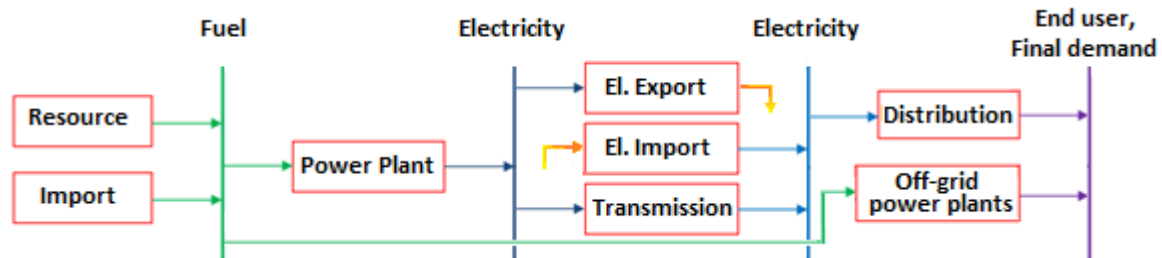


Figure 6: General Reference Energy System (RES). Note that also renewable resources such as water, sunlight and wind are counted as “fuels” in the models. These are not possible to import in the models used.

The total demand is specified yearly for the modelling period. Since demand and capacities are not always constant over a year, “time slices” divide each year into interesting periods for the modelling. The final electricity demand is the electricity that is to be consumed by end-users, for example industry. The electricity is provided to the users by transmission and distribution systems, which includes some losses and variable costs to be specified for both technologies. When off-grid power plants are included in the model, these are spared the transmission and distribution losses since off-grid power plants are assumed to be directly connected to the end-user. Also the different power plants are assigned different properties by the use of the parameters listed in Table 1 in paragraph 2.1.4. Some parameters, for example the capacity factor, are defined for each time slice. Good examples of generation technologies where the capacity factors are having big variations over the year are PV panels without storage, which cannot convert power during night time.

All power plants are running on fuels, i.e. also sun, wind and water are counted as such in the models. In some cases, different power plants use the same type of fuel, and in a few cases one power plant can use two different fuels. Different power plants can use the same or different fuels. The fuel producing technologies, i.e. the extraction/refining (“Resource” in Figure 6) or import of fuels, are linked to different fuel costs. Those can either be assumed to be constant or varying over the modelling period. The resource technologies do not consider any losses in the fuel production, but some fuels/technologies are limited either by a maximum annual stock, maximum capacity or maximum power output depending on the data available. This is further explained in paragraph 2.2.4.

Below, the main technologies used are shortly described. For each model, one national transmission system and three distribution systems for the different demands are used.

|                                                   |                                                                                                                                                                                                                      |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Distribution to Industry</b>                   | <i>System that distribute transmitted electricity to industry</i>                                                                                                                                                    |
| <b>Distribution to Rural (or Other consumers)</b> | <i>System that distribute transmitted electricity to rural areas. When the separation Industry and Other consumers is used, data for rural distribution costs are applied on Other consumers in the models used.</i> |
| <b>Distribution to Urban</b>                      | <i>System that distribute transmitted electricity to urban areas</i>                                                                                                                                                 |
| <b>Transmission</b>                               | <i>Transmission system for domestically generated electricity</i>                                                                                                                                                    |

When data is not available for all three demands, the demands as well as the distribution systems are instead separated into Industry and Other consumers. Two of the countries investigated, Egypt and Zimbabwe, are importing or exporting electricity. In the models, each country has one import/export technology for existing and/or planned country interconnections.

|                       |                                                                                                                                                                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Import /Export</b> | <i>Imported foreign electricity or export of domestic electricity. One Import/Export technology is used for each new or existing connection, i.e. a country that is connected to for example three foreign grids will have three import/export technologies.</i> |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Several power generating technologies are used in the models. However, all technologies are not made available in all models. For example it is assumed that nuclear power plants will not be built in a country within the model period if detailed construction plans are not made today.

|                     |                                                                                                                                                                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bagasse CHP</b>  | <i>Combined heat and power plant running on bagasse only. All plants using only bagasse are assumed to be of this plant type. Except fuel costs, parameters are the same as for Biomass CHP. Only applied in the Mauritius model.</i> |
| <b>Biomass/Coal</b> | <i>Direct co-firing with bagasse and coal. Assumed to use 50 % biomass and 50 % coal. Only applied on the Mauritius model.</i>                                                                                                        |
| <b>Biomass CHP</b>  | <i>Combined heat and power plant running on biomass. All plants running on biomass are assumed to be of this plant type.</i>                                                                                                          |
| <b>Biogas CHP</b>   | <i>Combined heat and power plant running on biogas/landfill gas. All plants running on biogas/landfill gas are assumed to be of this plant type.</i>                                                                                  |
| <b>Coal STPP</b>    | <i>Steam power plant running only on coal. All plants using only coal are assumed to be of this plant type.</i>                                                                                                                       |
| <b>Diesel DE</b>    | <i>Diesel engine (internal combustion) running on diesel.</i>                                                                                                                                                                         |

**Diesel DE, decentralized**

*Decentralized small diesel engine (internal combustion) running on diesel. The decentralized diesel engines have different parameters depending on if the diesel engine is used in industry or not.*

|                                 |                                                                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HFO OCGT</b>                 | <i>Open/single cycle gas turbine running on heavy fuel oil.</i>                                                                                   |
| <b>Hydro, small</b>             | <i>Conventional hydro power plant with a capacity less than 1 MW. Used for off-grid hydro and when plant specific data is missing.</i>            |
| <b>Hydro, medium</b>            | <i>Conventional hydro power plant with a capacity between 1-10 MW. Used when plant specific data is missing.</i>                                  |
| <b>Hydro, large</b>             | <i>Conventional hydro power plant with a capacity larger than 10 MW. Used when plant specific data is missing.</i>                                |
| <b>Site specific hydro</b>      | <i>Site specific data for hydro power plants is used when available.</i>                                                                          |
| <b>NG OCGT</b>                  | <i>Open/single cycle gas turbine running on natural gas. Also include all plants running on natural gas except CCGT, STPP and diesel engines.</i> |
| <b>NG CCGT</b>                  | <i>Combined cycle gas turbine running on natural gas.</i>                                                                                         |
| <b>Nuclear</b>                  | <i>Nuclear power plant.</i>                                                                                                                       |
| <b>CSP</b>                      | <i>Concentrated solar power plant.</i>                                                                                                            |
| <b>CSP storage</b>              | <i>Concentrated solar power plant with the possibility to store energy.</i>                                                                       |
| <b>CSP with gas firing</b>      | <i>Concentrated solar power plant with gas firing (natural gas).</i>                                                                              |
| <b>PV utility</b>               | <i>Photovoltaic solar power plant, utility scale without storage.</i>                                                                             |
| <b>PV rooftop</b>               | <i>Photovoltaic solar power plant, residential scale without storage.</i>                                                                         |
| <b>PV rooftop, 1 hr storage</b> | <i>Photovoltaic solar power plant, residential scale with battery storage, 1 hr.</i>                                                              |
| <b>PV rooftop, 2 hr storage</b> | <i>Photovoltaic solar power plant, residential scale with battery storage, 2 hr.</i>                                                              |
| <b>Wind 25 %</b>                | <i>Wind turbine (HAWT) with a capacity factor of 25 %.</i>                                                                                        |
| <b>Wind 30 %</b>                | <i>Wind turbine (HAWT) with a capacity factor of 30 %.</i>                                                                                        |

To limit the complexity of the models, the number of fuels used is limited. To make this possible, some fuels are assumed to include some other fuels with similar characteristics.

|                |                                                                                                                                                                                               |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Diesel</b>  | <i>Diesel fuel, also includes light fuel oil (LFO), gasoil, naphtha, oil distillate and unspecified oil/petroleum fuels</i>                                                                   |
| <b>HFO</b>     | <i>Heavy fuel oil, also includes crude and residual oil</i>                                                                                                                                   |
| <b>NG</b>      | <i>Natural gas, also includes methane</i>                                                                                                                                                     |
| <b>Bagasse</b> | <i>Residual product from sugar industry (Mauritius). Assumed to not have any variable cost, since many of the bagasse and bagasse/coal power plants are run by the sugar industry itself.</i> |
| <b>Biogas</b>  | <i>Biogas/landfill gas</i>                                                                                                                                                                    |
| <b>Biomass</b> | <i>Biomass, includes waste</i>                                                                                                                                                                |
| <b>Coal</b>    | <i>Coal, unspecified type</i>                                                                                                                                                                 |

|                    |                              |
|--------------------|------------------------------|
| <b>Solar</b>       | <i>Solar radiation</i>       |
| <b>Wind</b>        | <i>Wind</i>                  |
| <b>Uranium</b>     | <i>Uranium, nuclear fuel</i> |
| <b>Electricity</b> | <i>Imported electricity</i>  |

#### 2.1.4 Parameters used

To model the energy carriers (fuels) and technologies several parameters are used in OSeMOSYS. Below in Table 1 all parameters used in this project are described. The parameters could either be time dependent or time independent.

**Table 1: Parameters and designations used in the models, for time dependent parameters the column "A/TS" represent if the parameter is specified per year (A) or per time slice (TS)**

| Name                             | A/TS | Unit                     | Description                                                                                                                            | Use in model                                                                                                                                                   |
|----------------------------------|------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Time dependent parameters</b> |      |                          |                                                                                                                                        |                                                                                                                                                                |
| <b>Availability Factor</b>       | A    | -<br>(value between 0-1) | Gives the ratio between the time when the technology is available and the total time of a period.                                      | To model the availability of technologies                                                                                                                      |
| <b>Capacity Factor</b>           | TS   | -<br>(value between 0-1) | The capacity factor gives the ratio between the real output of a technology and the output of the same technology working 8760 h/year. | To model the capacity factor of technologies                                                                                                                   |
| <b>Fixed Cost</b>                | A    | mUSD/GW                  | Costs connected to the installed capacity of a specific technology and year                                                            | Fixed O&M costs (sometimes including variable O&M costs)                                                                                                       |
| <b>Input Activity Ratio</b>      | A    | -                        | Gives the ratio between the input of an energy carrier/fuel and the activity of the technology.                                        | Together with Output Activity Ratio it represents efficiency of a technology.                                                                                  |
| <b>Output Activity Ratio</b>     | A    | -                        | Gives the ratio between the output of an energy carrier/fuel and the activity of the technology.                                       | Used with Input Activity Ratio to represent the efficiency of a technology. In most cases set to 1.                                                            |
| <b>Residual Capacity</b>         | A    | GW                       | The capacity that is already installed when the model period starts.                                                                   | Used to model technologies put into use before 2011. The value is constant or more often decreasing over the time period as plants are supposed to be retired. |
| <b>Specified Annual Demand</b>   | A    | PJ                       | The demand for a specified energy and year.                                                                                            | Represents the projected electricity                                                                                                                           |

|                                                     |                          |                          |                                                                                                   |                                                                                                                    |
|-----------------------------------------------------|--------------------------|--------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|                                                     |                          |                          |                                                                                                   | demand, in the model separated in different sectors.                                                               |
| <b>Specified Demand Profile</b>                     | TS                       | -<br>(value between 0-1) | Defines the share of the Specified Annual Demand that belongs to a specific time slice.           | Used to model how the demand varies over a year.                                                                   |
| <b>Total Annual Max Capacity Investment</b>         | A                        | GW                       | The maxim capacity of a specific technology that can be invested in a specific year.              | Used to limit the yearly investments in a technology.                                                              |
| <b>Total Annual Min Capacity Investment</b>         | A                        | GW                       | The minimum capacity of a specific technology that is to be invested in for a specific year.      | Used to model planned investments in specific technologies.                                                        |
| <b>Total Technology Annual Activity Upper Limit</b> | A                        | PJ                       | The maximum amount of energy that can be produced by a specific technology a specific year.       | Used to model limited resources.                                                                                   |
| <b>Variable cost</b>                                | A                        | mUSD/PJ                  | Costs connected to the energy output of a technology.                                             | Variable O&M costs (sometimes including fixed O&M costs)                                                           |
| <b>Time independent parameters and designations</b> |                          |                          |                                                                                                   |                                                                                                                    |
| <b>Capacity To Activity Unit</b>                    | -                        |                          | The relation between the units for capacity and activity.                                         | Since the capacity and energy in this model are GW and PJ respectively, the factor is set to 31.536.               |
| <b>Discount Rate</b>                                | -                        |                          | Gives the discount rate that should be used for a specific technology. Used to calculate the NPV. | In this model, all technologies are assumed to have the same discount rate, 10 %.                                  |
| <b>Operational Life</b>                             | Years                    |                          | The lifetime of a technology, not specified for residual capacities.                              | Gives the maximum time a new technology can be used without new investments.                                       |
| <b>Time Slice</b>                                   | Not a parameter (!)      |                          | An optional number of time slices can be used to subdivide every year into smaller parts.         | Used to model how for example the demand and capacity factors vary over the hours of a day and/or between seasons. |
| <b>Year Split</b>                                   | -<br>(value between 0-1) |                          | The share of a year that a specific time slice represents.                                        | Used to model the length of the different periods (time slices) of the year.                                       |

## 2.2 Model assumptions

Several assumptions are used in the OSeMOSYS models. In this chapter a summary of the most important ones will be presented.

### 2.2.1 Overall assumptions

The following assumptions describe the general context in which the model will undergo the optimization process. These are flexible and can be modified to fit different output specifications. In the data set-up used in this project, the following assumptions are made.

- The discount rate is set to 10 %.
- The monetary unit is 2010 USD.
- The study horizon spans from 2010-2030, with focus on 2014-2020.
- The models can only invest in off-grid alternatives (diesel, PV and small hydro) until 2014, the following years the technology investments are unlimited. PV, CSP, hydro and in the Mauritius model also bagasse are however limited due to limited resources.
- The year is split into twelve periods; one division is used for islands (Madagascar and Mauritius) and one for inland countries (Egypt and Zimbabwe), see *Appendix C- Country specific data*, Paragraph C.1, for detailed data.
- Electricity demand and some technology characteristics, as for example bagasse or solar availability, vary over those periods.
- The yearly electricity demand is divided into three categories, namely Industry, Urban and Rural demand. When less data is available, the demand is divided into Industry and Other consumers.

### 2.2.2 Assumptions about electricity demand

The electricity demand is specified for each year 2010-2030 and is based on country specific forecasts. More information about this is to be found in *Appendix C-Country specific data*.

The yearly demand is separated into three sectors as listed below.

- Industry
- Urban
- Rural
- (Other consumers, used in addition to Industry only if Urban and Rural data is not available)

In the models of Madagascar, Mauritius and Zimbabwe the preferred split was not available, which is why the second choice Industry and Other consumers is used. The shares of the total demand that was connected to the different demands are assumed to be constant during the whole modelling period. Since the different demands do not have any other sector specific attributes (as for example different distribution losses) this division is only made to indicate how the demand is built. Except the use of off-grid power plants the split does not affect the simulation.

To simulate how the demand varies over days and season, each year is, as mentioned in paragraph 2.1.3, separated into different time slices. In this project, two different set ups of time slices are used. The islands models in this project, Mauritius and Madagascar, are based on twelve time slices considering the difference between daytime, night and peak demand (evening), weekends and workdays as well as two seasons, summer and winter. Daytime is 06:00-17:00, peak 17:00-23:00 and night 23:00-06:00. For the inland countries, Egypt and Zimbabwe, the day is divided into three parts where day Part 1 consists of 12 hours, Part 2 of 5 hours and Part 3 of 7 hours. Here, four seasons are used. Season 1 is based on data for January, Season 2 is based on April data, Season 3 is based on July data and Season 4 on October data. The four seasons does not have equal lengths. More details on the time split can be found in *Appendix C – Country specific data*, paragraph C.1.

For the Egypt model, all demands were provided by KTH-dESA (dESA, 2014). For the remaining models, the demands for 2010 are from African Regional Energy Statistics (AfDB Group, 2013).

The demand forecasts are done in different ways for the four models, depending on what data was available.

- **Egypt:** Data from KTH-dESA is used.
- **Madagascar:** The forecast is based on predicted peak demands for Antananarivo grid 2011-2020 (ECFA, 2009). From 2020 the annual perceptual growth is assumed to be constant and have the same value as 2020.
- **Mauritius:** Two electricity demand growth predictions from CEB (Central Electricity Board, 2012) one total and one for industry, are used. The two forecasts are for the years 2011-2022 but the mean perceptual growth is used for the whole modelling period.
- **Zimbabwe:** Historical data for the GDP 2005-2014 (AfDB Group, 2014b) is used to extract a mean percentage annual growth. This is then applied on the whole modelling period.

### 2.2.3 Assumptions on transmission and distribution

One country is rarely similar to another when it comes to transmission and distribution systems. In this project, the transmission and distribution systems are very simplified. The potential existence of small grids is neglected and all power plants in a country are assumed to be either off-grid or connected to one national grid with the same transmission losses everywhere, independent of for example the voltage or the distance between the power plant and the distribution system. For import and export of electricity, losses are neglected. Even though imported electricity in reality are connected to national grids, it is not connected to the transmission systems in the models. Parameters, such as capital cost and availability, are instead put directly on the import/export technology.

Each demand, i.e. Industry, Urban and rural (or Industry and Other consumers), is assigned a distribution system. Those are assumed to be used only to distribute power from grid connected plants. New off-grid technologies are assumed to not have transmission or distribution losses. Data used regarding costs and other parameters of each country's transmission and distribution systems are presented in *Appendix B-Default data*.

Future import or export of electricity between islands is not considered as a likely scenario because of the shielded location of the countries.

### 2.2.4 Assumptions on existing and planned capacity

The existing capacity in the models is based on PLATT-data (WEPP, 2013). PLATT gives information about who owns the power plant, where it is located, how much capacity is installed, when it was introduced and when some of the new power plants will be introduced.

Existing and planned hydro power plants are modelled separately, whereas other technologies are modelled as "groups" of power plants, including all capacity of the technology in question. This is due to the fact that hydropower parameters are depending on the size and location of the power plant. Even if this could be the case also for other technologies, the remaining technologies are grouped in order to simplify the models. All technologies used are listed in *Appendix C- Country specific data*. When plant specific data is not available, general data for hydropower is used, considering three plant size intervals.

The countries that have been studied in this project have different natural resources that can be used in electricity production. Many of the countries rely on fuel imports. The OSeMOSYS model takes into account that the resources can either be locally produced, if possible, or imported. The import fuel price is assumed to be 10 % higher than the cost of domestic extraction.

For the countries that are assumed to have domestic resources, some resources are limited. Potential PV, CSP and wind capacities are limited based on data from KTH-dESA. Off-grid hydro potential in each

country is based on data from the small hydropower data portal Small Hydro World (UNIDO and ICSHP, 2014). Grid connected hydro power is assumed to be built only if planned.

### **2.2.5 Variations in capacity**

It can be assumed that the capacity factors of several technologies, as for example wind turbines, solar technologies and hydropower, are fluctuating over the year due to dry/rainy seasons etc. In the models presented in this report, seasonal variations are neglected due to limited data. However, solar technologies still have variations over the day. For PV/CSP technologies without storage, the generation is assumed to be limited to daytime. For technologies with storage options the capacity factors are reduced during evenings and nights. When the data is available, the models can easily be modified in order to be more realistic.

Regarding wind power, it is assumed that all wind turbines are on-shore with a capacity factor of 25 % and/or 30 %. In this initial version of the models, all existing and planned wind turbines are assumed to have a capacity factor of 30 %. Capacity factors for all technologies, as well as other relevant data, are presented in *Appendix B – Default data* and *Appendix C – Country specific parameters*.



## 3 RESULTS

All results presented here correspond to output data from modelling in OSeMOSYS. The results are based on a Baseline scenario and are presented for year 2010 – 2025, since the results of the last years of the modelling period are irregular and unreliable due to the setup of the software. In the last years of modelling, an “edge effect” occurs relating to how investments are managed in the equations building the software. All results show a significant change 2014, due to the constraints in technology investment that are released this year.

The four modelled countries has different preconditions, for example the power demand 2010 in Egypt is about 100 times larger than the demand in Mauritius the same year. For all models, hydropower and coal STPP are frequently seen in the projected electricity mixes. In Madagascar and Zimbabwe, the mentioned technologies together with diesel and HFO plants make most of the installed capacity and power generation. Egypt differs from the other countries due to their high capacity and use of natural gas technologies. Mauritius also stands out, due to their high use of off-grid PV.

### 3.1 Egypt

The installed capacity in Egypt increases from 35.7 GW 2010 to 86.6 GW 2025. As seen in Figure 7, the model chooses to invest in 2.8 GW of CSP with gas-firing (CSP/Natural Gas), mainly year 2014, i.e. the first year that the model is free to make new unlimited investments. From this year, the increasing demand is mainly met by new installations of natural gas power plants. The majority of the new natural gas plants are of the type CCGT. From 2018, this is the only natural gas technology invested in. The total natural gas capacity (not including CSP with gas-firing) increases from 26.3 GW 2010 to 64.3 GW 2025. Year 2020 the model invests in a 1.0 GW nuclear power plant. Minor investments are also made in on-grid diesel engines, hydropower and wind turbines as well as off- and on grid PV. Since the beginning of the modelling period there are installed capacities of biomass CHP (0.1 GW) and HFO OCGT (2.1 GW), but the capacities of these are decreasing do to the absence of new investments and the assumed retirements of individual plants.

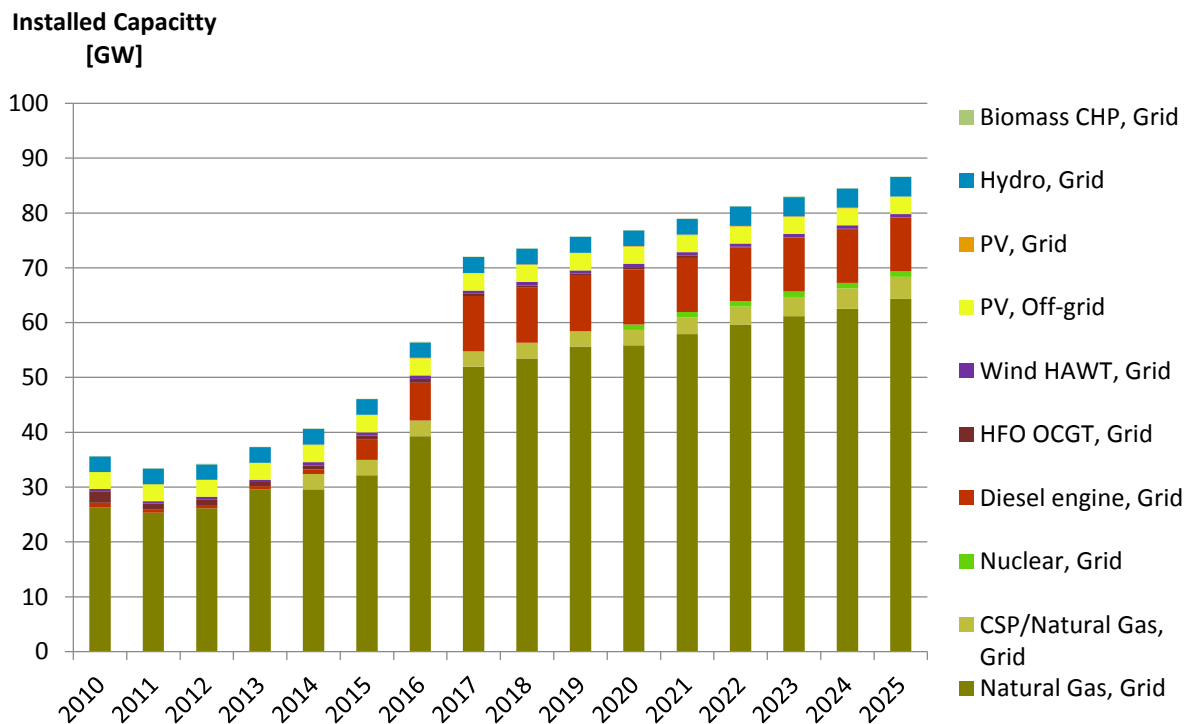


Figure 7: Total installed capacity in Egypt

According to the results presented in Figure 8, the electricity in Egypt will mainly be produced by technologies running on natural gas. Alone, natural gas plants generated 509.9 PJ in 2010, compared to the total power generation of 571.5 PJ (89 %). The same year, the second largest power technology is hydro power, generating 39.2 PJ followed by off-grid PV generating 17.3 PJ.

2014, CSP with gas-firing starts to make a significant share of the electricity generation, 70.9 PJ, increasing to 101.1 PJ by the end of the presented time period. The generation from hydropower increases from 39.6 PJ annually 2010-2021 to 48.6 PJ annually 2022-2025. Wind power generation increases from 3.4 PJ 2010 to 4.9 PJ 2025. The new nuclear plant starts to run 2020 and are annually generating 26.8 PJ throughout the remaining years. However, the mentioned increases in capacities are still not close to the energy quantities generated by natural gas.

The existing capacities of petroleum technologies are not used after 2014. Biomass CHP is used throughout the modelling period, generating 0.7 PJ 2010 and 0.5 PJ 2025. 2025 the total power generation is 1175.8 PJ, with natural gas plants (CSP with gas-firing not included) making 83 % of the power.

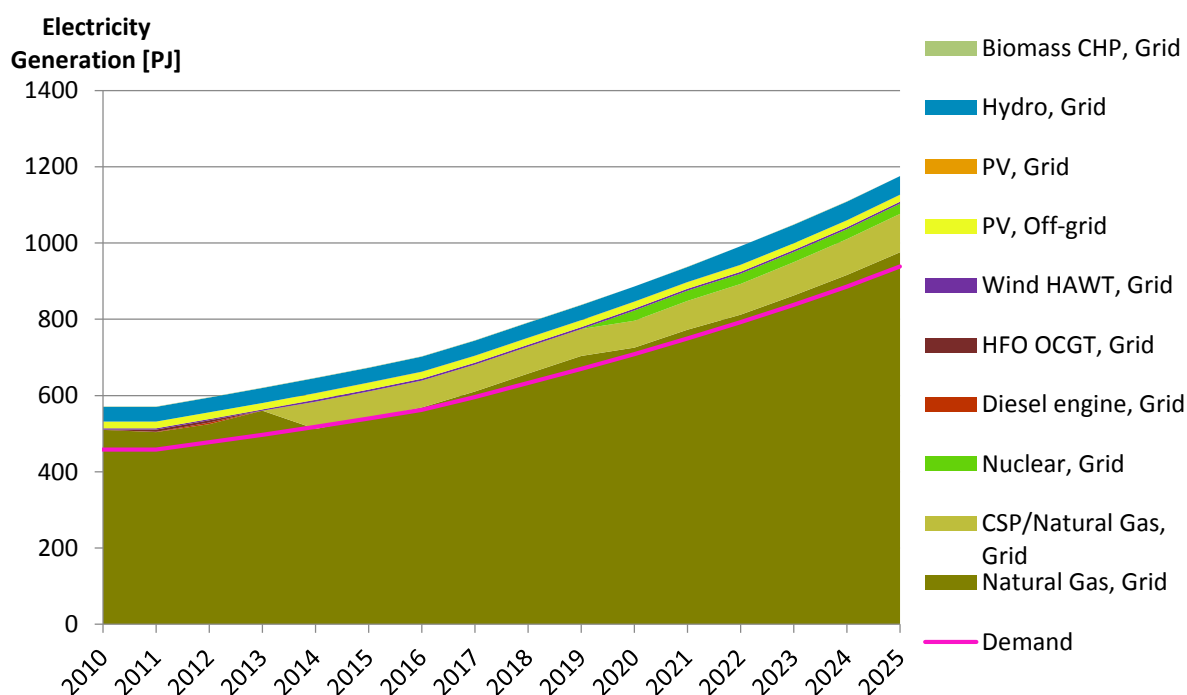


Figure 8: Egypt's power generation by technology

### 3.2 Madagascar

The installed power capacity in Madagascar is 0.5 GW year 2010, increasing to 1.1 GW 2025. As seen in Figure 9, the majority of the power plants installed are fossil fuel based throughout the modelling period. The installed capacity of grid-connected diesel engines decreases slightly from 0.3 GW 2010 to 0.2 GW 2025. Also HFO OCGT decreases slightly, even though this is barely visible in the figure. After the introduction of 0.2 GW coal STPP 2011, the installed capacity of this technology increases to 0.5 GW 2025. The renewable technologies installed are on- and off-grid hydropower and on-grid PV. Grid connected hydro increases from 0.1 GW 2010 to 0.2 GW 2025, while off-grid hydro, evenly split between industry and other users, stays constant at 20 MW (0.02 GW) after investments 2010. The last three years presented, investments are done in PVs. The investments 2023 are small, only 4 MW, but the following two years the installed capacity increases rapidly to 0.1 GW.

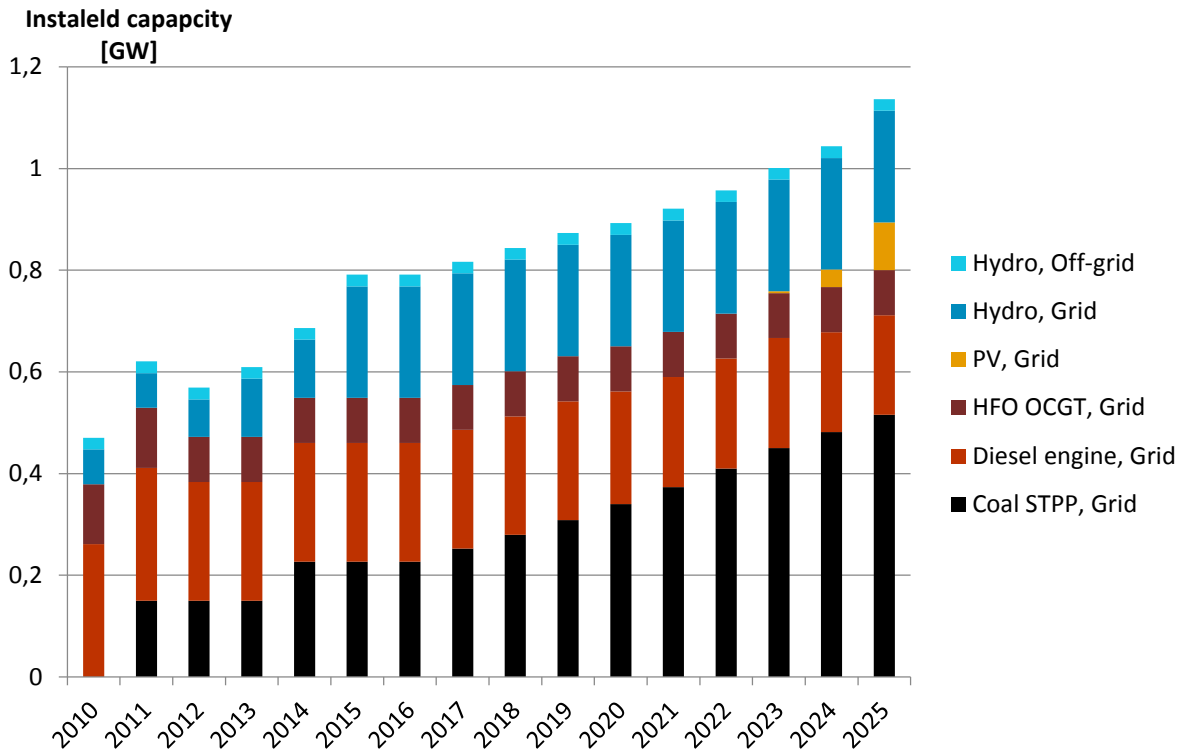


Figure 9: Total installed capacity in Madagascar

Figure 10 shows the power generation by technology in Madagascar. The technologies used are on- and off-grid hydropower, coal STPP and initially also HFO and diesel engines. Diesel engines and HFO OCGT together makes 89 % of the generated power 2010, with 1.7 PJ from diesel and 3.0 PJ from HFO. Diesel engines are only used 2010 and HFO OCGT is not used after 2014. Instead, coal STPP becomes the major technology, starting with 3.8 PJ 2011 and increasing to 12.9 PJ 2025. On-grid hydropower increases from 1 PJ 2010 to 3.1 PJ 2015 and is after generating the same amount of electricity per year. Off-grid hydro is fully utilized from 2010, generating 0.4 PJ yearly except 2025 when the generation decreases to 0.2 PJ.

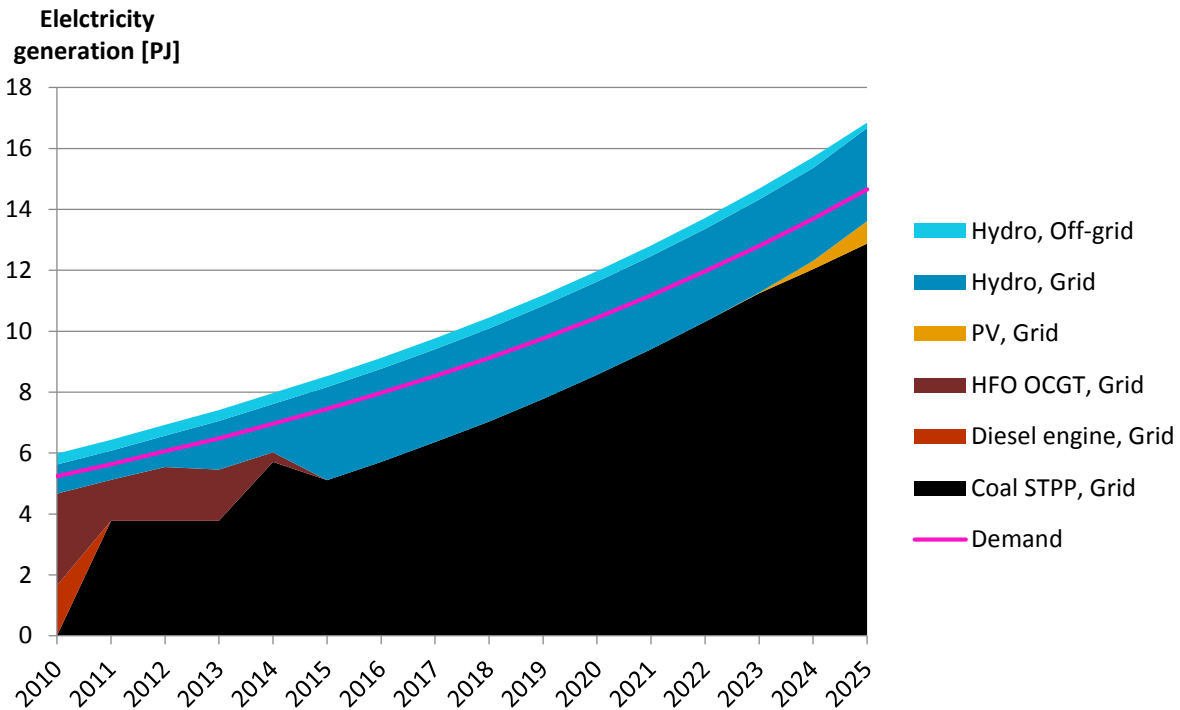


Figure 10: Madagascar's power generation by technology

### 3.3 Mauritius

The total installed power generation capacity in Mauritius is presented in Figure 11. Between 2010 and 2011 the capacity decreases from 1.8 GW to 1.5 GW due to assumptions made on the average plant lifetime. Several of the plants which run 2010 are old and have, according to the lifetime assumptions, already passed their retirement year. In the model, these are taken out of service, which explains the relatively large capacity drop. After 2011, the total capacity increases slightly up to 1.7 GW 2025. As can be seen in Figure 12, the capacity is still enough to meet the power demand.

Planned capacity investments in the Mauritius model include the technologies coal STPP and HFO OCGT as well as regular investments in off-grid PV and HAWT. However, the large investment in off-grid PVs 2010 is not planned. The results give that Mauritius invests in 0.5 GW off-grid PV (rooftop) 2010, staying almost constant until 2025. Mauritius has 10 MW (0.01 GW) grid-connected solar power 2010 which increases to 0.1 GW by 2025. The installed capacity of HFO OCGT decreases from 0.5 GW 2010 to 0.1 GW 2025. The installed capacity of diesel engines also decreases, starting at 0.1 GW 2010 and disappearing 2020. Regarding hydro power, Mauritius invests in 10 MW off-grid hydropower 2010, in addition to the existing on-grid capacity of 0.3 GW. Both capacities stay constant until 2025. 2015 and 2016 Mauritius has planned to install in total 0.1 GW of coal STPP. The investments does however not stay there, and 2025 the capacity has increased to 0.3 GW.

Bagasse is a sugar industry rest product which today (2014) contributes to a significant share of the total power generation in Mauritius. Fuel availability is linked to the yearly crop season, why several plants switch to coal fuel when the bagasse availability is low. No new investments are made in bagasse technologies; the capacity of plants running on bagasse instead decreases from 0.2 GW to 2 MW 2025 and plants with coal co-firing decreases slightly from the 0.2 GW 2010. Mauritius also has wind power, 2010 the capacity is 0.2 MW (0.0002 GW) but 2025 the capacity has increased significantly to 0.1 GW. Biogas CHP is introduced 2012 and has a capacity of 3 MW (0.003 GW), which stays constant until 2025.

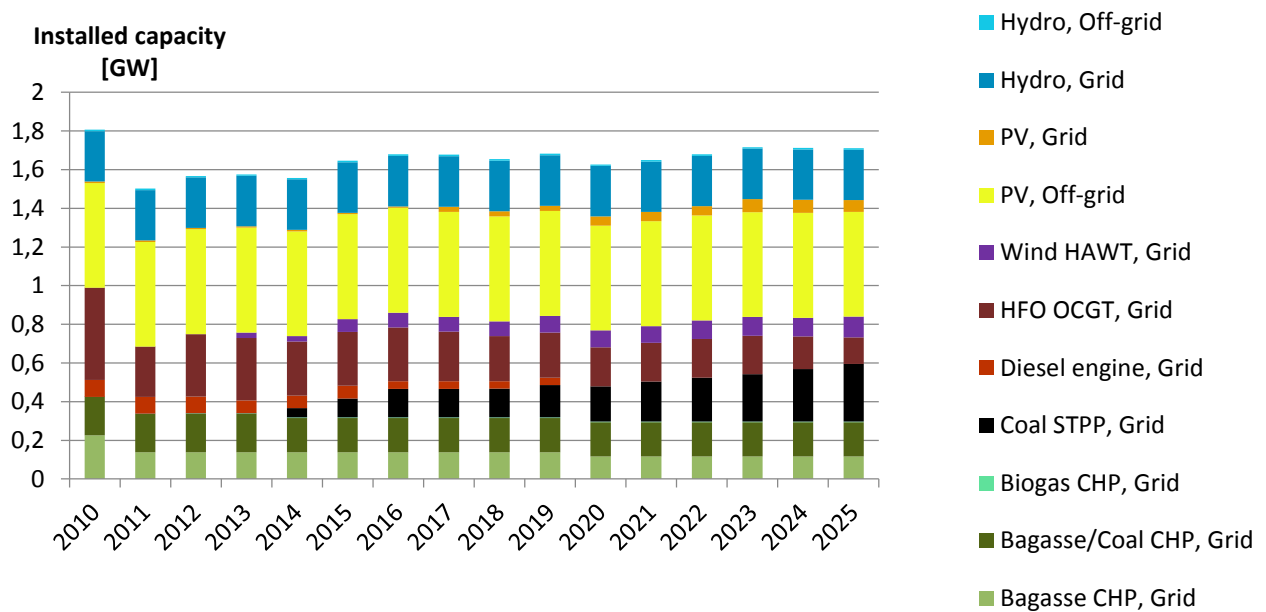


Figure 11: the total installed capacity in Mauritius

Figure 12 shows the annual power generation in Mauritius. As seen in the figure, the country immediately starts to utilize the large off-grid PV investments made 2010. After, the demand increase is mostly met by coal power. The total power generation increases from 9.5 PJ 2010 to 16.2 PJ 2025. The existing grid connected hydropower plants are fully utilized generating 3.6 PJ annually. Off-grid hydro is generating a constant amount of power annually, 0.1 PJ. Off-grid PV power generation also stays constant at 3.4 PJ annually. The power generation from grid connected PV is 0.1 PJ 2010 and increases to 0.5 PJ 2025. Bagasse based generation is limited by the annual stock. The total generation from bagasse and bagasse/coal plants stays around 1.7 PJ annually, even though the share of power from each of the both technologies varies.

Diesel engines and HFO OCGT only generates electricity during the first five years of the modelling period, with in total 0.6 PJ 2010, 1.4 PJ 2013 and 0.9 PJ 2014. Of the two, HFO OCGT is the by far largest power source as can be seen in the figure below. After the introduction of coal STPP, the share of power from the technology increases rapidly. 2014 coal STPP generates 1.0 PJ, 2025 it generates 6.0 PJ. Also generation from wind power (HAWT) advances, starting at 2 TJ 2010 (0.002 PJ) and increasing to 0.9 PJ 2025.

Biogas CHP is introduced 2012, generating 0.1 PJ, but is only used 2012-2014.

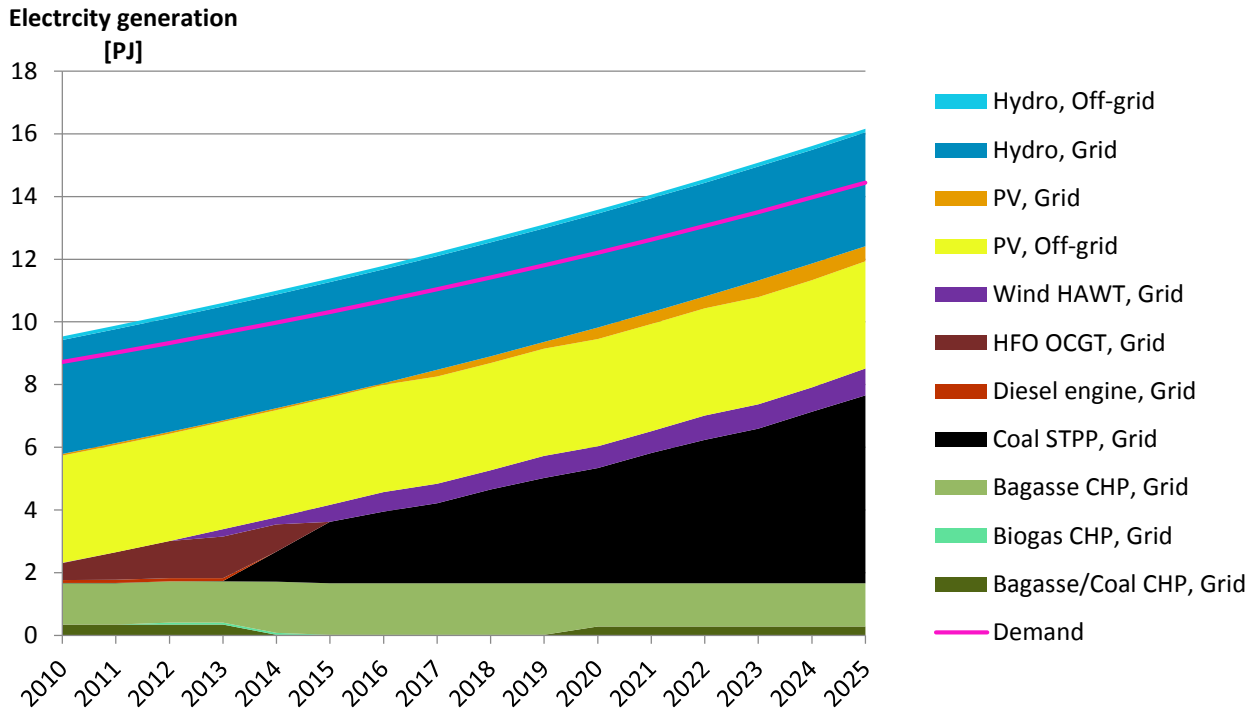


Figure 12: Mauritius's power generation by technology

### 3.4 Zimbabwe

The total installed power capacity in Zimbabwe starts at 2.8 GW 2010 and increases to 7.3 GW 2025. Coal STPP is the most common technology with 1.1 GW 2010 and 4.0 GW 2025. As seen in Figure 13 below, there is a dip in the coal capacity 2014. This is caused by assumed retirements. Grid connected diesel engines and hydropower plants also makes significant shares of the installed capacity, with 0.5 and 0.7 GW respectively 2010 and 1.5 and 1.3 GW 2025. The import capacity stays constant over the modelling period at 0.3 GW. Year 2010 the model invests in 0.1 GW of off-grid hydropower for industrial use. Even this capacity stays constant. The existing capacity of 0.1 GW biomass CHP is decreasing slightly, once again due to assumed retirements.

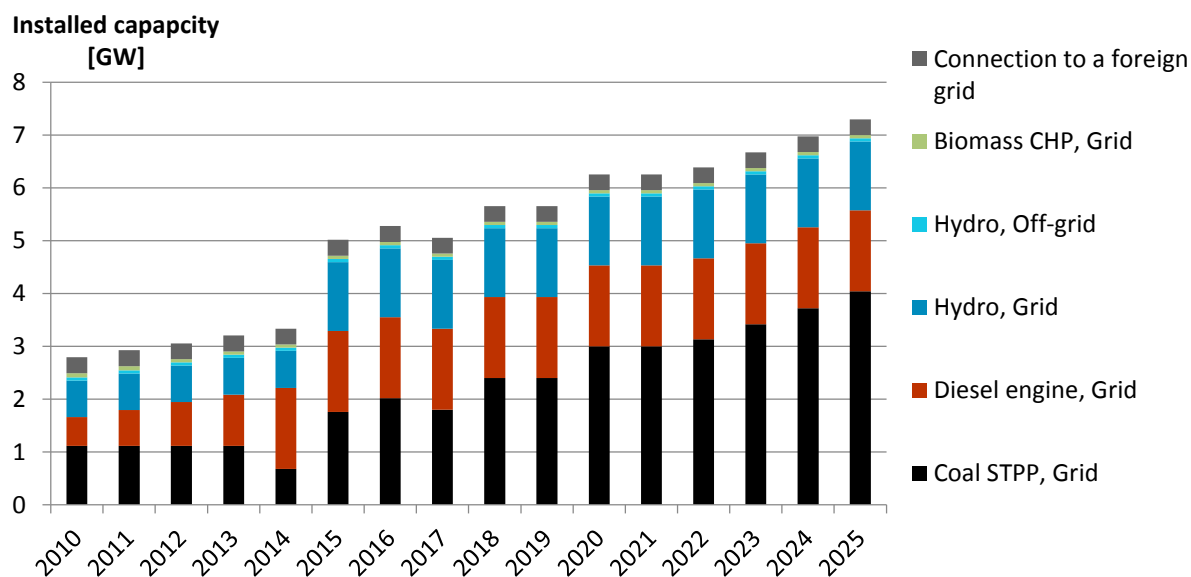


Figure 13: the total installed capacity in Zimbabwe

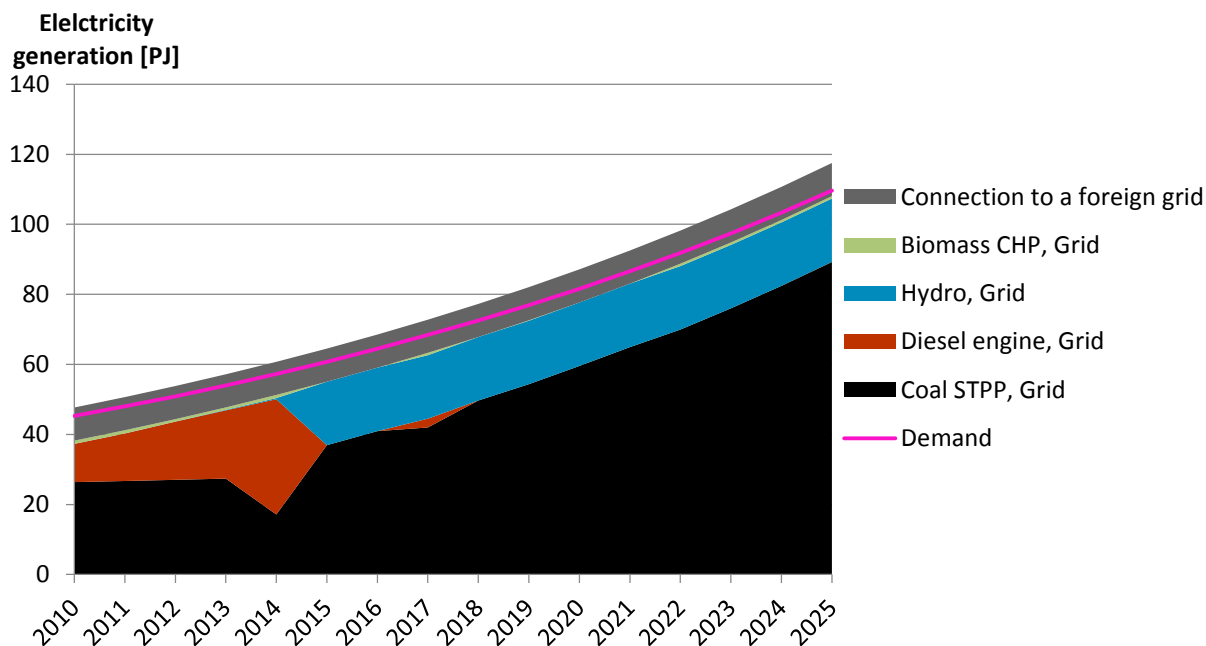


Figure 14 demonstrates the results for the present and future electricity generation in Zimbabwe. The generated power is more than doubled over the presented years, increasing from 48.6 PJ 2010 to 118.4 PJ 2025. Coal is the dominating power source, and the share of electricity generated using the fuel is steadily increasing except a dip 2014, which is caused by the capacity dip the same year. Coal STPP generates 26.4 PJ electricity 2010 and 89.2 PJ 2025. Hydropower makes a small share of the electricity in the first years; 2010 grid connected hydropower generates only 30 MW (0.03 PJ) and off-grid hydro 0.9 PJ (for industrial use). Due to new investments in grid-connected hydropower plants, the on-grid hydro increases to 18.1 PJ 2025 while the off-grid plants stay at a constant level. The imported electricity is fully utilized, 9.5 PJ, during the whole modelling period. Grid connected diesel engines are used mainly during the first five years, starting at 10.9 PJ 2010 and peaking at 32.9 PJ 2014. Biomass CHP is used for power generation most of the simulated years, the amount of power fluctuates with values between 0.0 PJ and 0.9 PJ.

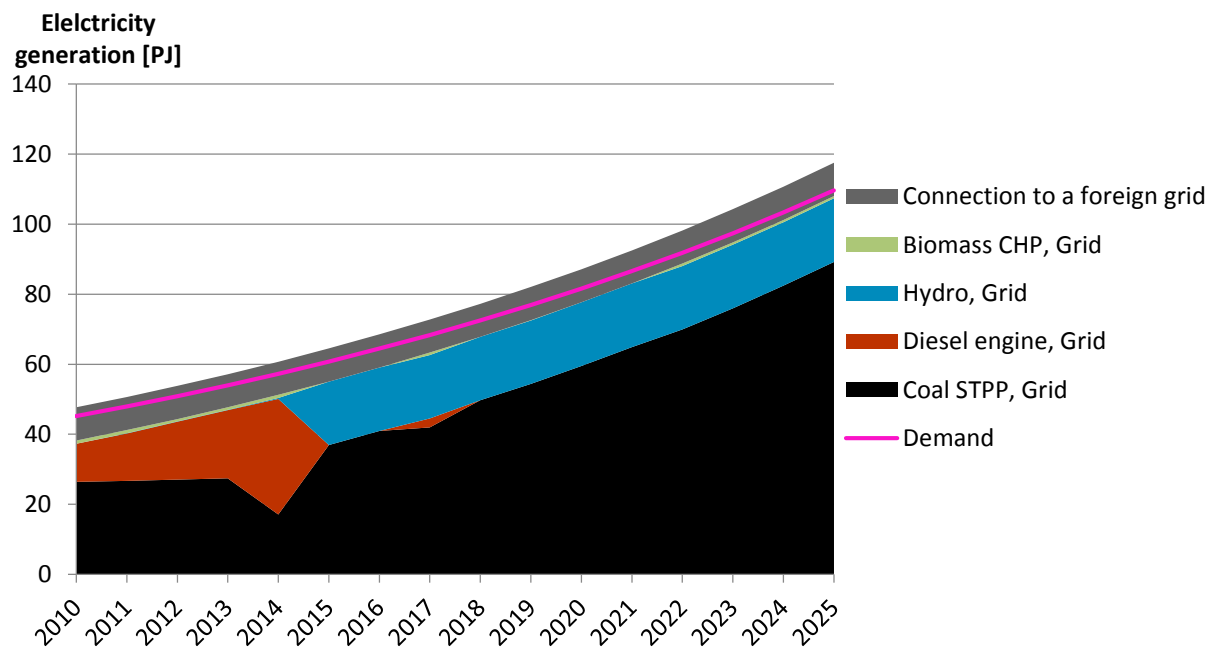


Figure 14: Zimbabwe's power generation by technology

## 4 DISCUSSION AND CONCLUSIONS

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This chapter includes a discussion about the work and the results that this master thesis has generated and offers some summarizing conclusions. Next step for the project between AfDB and KTH-dESA are also mentioned.

### 4.1 Discussion

Functional models of the electricity sectors of Egypt, Mauritius, Madagascar and Zimbabwe are produced. However, the models are to be re-populated with up-to-date data that is to be collected by AfDB. The existing results contain no evaluation of error margins. This, together with other factors mentioned in this paragraph, gives that the results presented is not yet ready to be used by policy makers. However, it can still be used as an indicator of what technologies could be interesting alternatives from an economic perspective. The created models will later be populated and run by AfDB and KTH-dESA, generating the results needed for the AfDB web-application. One single simulation per country model, as presented in this thesis, cannot be directly used in the web-application since it for each country needs results for all possible scenarios (as described in paragraph 2.1.2 and 4.3). The models are to be populated with data provided by AfDB why the data setup used in these models can be seen as temporary. Since the focus of this project has been to create models rather than provide results, less effort have been put on evaluating the results.

The models are made and evaluated in the software ANSWER OSeMOSYS. For more trustworthy results, the models could be built in several different programs in order to make sure that there are no software related errors. This has not been considered as necessary in this project.

The modelling period used (2010-2030) is confusing since the period starts four years in the past. It has not been possible to start the modelling at 2014 due to a lack of up-to-date data. There could also be advantages with starting the modelling period in the past, as for example the possibility to compare the results to measured data as well as the avoidance of potential edge problems. Edge problems are also the reason to the prolonged modelling period. In this project, the last ten years of the period is only simulated to avoid the sometimes irregular results that, due to software related issues, can occur especially in the end of a modelling period. However, the authors chose to present the results for 2020-2025 since the results are available and no irregularities seemed to occur within that interval. It could also be further evaluated how the length of the period affects the investments.

Since the models are driven by a final demand, the demand makes a big potential source of errors. In the models produced in this project the demands used can be contested. To make the result more reliable, more analysis needs to be done regarding the demand projections. Zimbabwe's demand are based on historical GDP increase, and it can be questioned if this is a valid assumption that the GDP will continue to follow the historical patterns during the rest of the modelling period and if GDP and electricity consumption are very related to each other. For all models, several demand projections should be used. This has not been done in this project due to time constraints and the fact that, as mentioned earlier, focus has been to create functional models and not results.

Regarding the results, there are both similarities and differences between the countries. Generally, hydropower, coal and natural gas seem to be feasible options when made available. It should however be clarified that the present model set-ups does not consider emissions or other environmental factors. Hydropower capacity is assumed to be limited to existing and planned plants, due to the fact that the modelling period of interest for AfDB, 2014-2025, is assumed to be too short for unplanned investments in large hydro projects. This could be adjusted based on more detailed information about potential capacity investments. The Mauritius model has a remarkably high share of off-grid PV. This is interesting but potentially also a bit unrealistic, considering that off-grid technologies often include higher costs than grid-connected options. Costs etc. of the off-grid rooftop PV without storage therefore

needs to be evaluated, in order to see if the result seems realistic. For all models, the technologies and fuels made available should also be assessed. It is possible that the results for Mauritius, as well as the remaining countries, will change when using the data provided by AfDB.

The modelling of the electricity imports and exports in Egypt and Zimbabwe could have been more detailed. In the present models there are no demands for electricity export, why the Egypt model is not trading any electricity. This is however easily adjusted if/when data is available. The model of Zimbabwe is the only of the two that imports electricity. It seems strange that the model chooses to utilize the import throughout the whole modelling period, since the length of the time span allows the model to do long term investments. This could be further evaluated in order to see if the result is unrealistic or not. It is assumed that no new import/exports will be done except existing and planned projects. The models could be adjusted so that unplanned connections are made available in the end of the modelling period.

In the present version of the models, the investments restrictions are loosened in 2014, which causes significant changes in the electricity mix. This could probably be done in a more realistic way, by investigating what technologies/knowledge is available in the respective countries and what time it would take to plan and construct different power plants. In future versions of the models, some investments should be restricted for a longer period. The reason to why the investments are loosened up so early is that the model period starts at 2010. A majority of the capacity data is from 2011 and can be considered as relatively old. The present assumptions in the modelling entails that a power plant that was planned and in the beginning of its construction 2011 potentially could be able to generate electricity 2014, even if this may be too optimistic in many cases.

The results could also be further evaluated. OSeMOSYS provides more information than the results presented here, for example regarding costs and variations over the different time slices. This was not considered as very relevant in this paper, since the two figures presented for each model were considered sufficient to give an overview of the results. Because of the unevaluated sources of errors in the models, it could also be considered to be excessive to present too detailed information.

When this project started in February 2014 it was assumed that AfDB had access to the raw data needed as well as detailed information about the web-application where the results of this project and the KTH-dESA – AfDB collaboration is to be used. This turned out to not be the case, which is why the scope of the project has been amended during the process. The major changes included limitations of the countries modelled as well as assisting the AfDB developing a form which is to be filled with data by people with knowledge about the countries of interest. It is unclear if these misunderstandings could have been avoided or solved during an earlier stage of the project. Better communication with AfDB could probably have made some details clearer in the beginning of the project period, even though problems along the process can be assumed to be a regular part of most projects. On a personal level, the authors of this paper found this project, including the procedures connected to it and the close collaboration with an international development organization, very educational.

## **4.2 Conclusions**

- The most commonly used power technologies in the countries, seen over the whole modelling period, are coal STPP, hydropower and natural gas (Egypt).
- Zimbabwe is the only country that imports electricity.
- The models produced in this project are “functional”, but the data needs to be evaluated in order to make the results more reliable. The models can also be adjusted, especially concerning investment limitations and electricity import/export.
- The remaining objectives of the project, i.e. the completion of two minor country reports and the AfDB input data table as well as establishing contacts at AfDB, was completed successfully but are not presented in this report.

### **4.3 Future work**

The collaboration between AfDB and KTH-dESA will continue during the autumn 2014. AfDB are collecting input data from the countries of interest, which later will be used by KTH-dESA to populate the OSeMOSYS models. Further discussions about the web application will also be held, in order for KTH-dESA to know what parameters they need to vary in the simulations and what intervals are of interest for these parameters. This will result in a comprehensive number of results, one for each possible combination of the variable parameters. These then need to be adjusted to a format suitable for the AfDB web-application.

In the future model set-up, more parameters will be considered. Environmental impacts, urban population growth etc. are important factors that most likely will be included in the final models.

Regarding the models, potential improvements for example include assumptions about the restrictions of technology investments. The models can also be more detailed and include several grids per country as well as more technologies and fuels. Electricity trade should be further evaluated and better implemented in the models. For the example hydropower, it can be distinguished between the type's run-of-river, dam and pumped storage, and the way neighbouring hydro power plants in some cases affect each other can also be modelled.



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## APPENDIX A – BASIC COUNTRY INFORMATION

In this appendix is the country information about the remaining fourteen countries presented.

### A.1 ALGERIA

|                                                        |                                               |                                                                                                                                     |
|--------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Official name: PEOPLE`S DEMOCRATIC REPOBLIC OF ALGERIA |                                               |                                                                                                                                     |
| Region:                                                | Northern Africa*                              |                                                  |
| Government type:                                       | Republic                                      |                                                                                                                                     |
| Currency:                                              | Algerian dinars (DZD)                         |                                                                                                                                     |
| Surface area, land:                                    | 23 817 41 km <sup>2</sup>                     |                                                                                                                                     |
| Population (2014 est.):                                | 38 813 722                                    |                                                                                                                                     |
| GDP/capita (2013 est.):                                | 7 500 USD**                                   |                                                                                                                                     |
| Exports                                                |                                               |                                                                                                                                     |
| - Total (2013 est.):                                   | 68.25 billion USD                             |                                                                                                                                     |
| - Commodities:                                         | Petroleum, natural gas and petroleum products |                                                                                                                                     |
| Imports                                                |                                               |                                                                                                                                     |
| - Total (2013 est.):                                   | 55.02 billion USD                             |                                                                                                                                     |
| - Commodities:                                         | Capital goods, foodstuffs, consumer goods     | Unspecified references (including map): (CIA, 2014)<br>*Geographical region by (UN Statistics Division, 2013)<br>**Data in 2013 USD |

Algeria is Africa's biggest country and located in the northern Africa. Algeria has a coastline in north and bordered by Mali, Nigeria, Libya, Morocco, Mauritania and Western Sahara. The climate in north Algeria is quite hot and dry during summer season and wet during winter, further south is the desert with extremely dry and hot summers and mild winters (CIA, 2014). Algeria is considered to be an upper middle income country (World Bank, 2014a).

The total primary energy supply was 2009 about 1 664.6 PJ, of which biomass represented 0.1 %, petroleum products 40 % and hydro 0.1 %, Coal and Coal products 1 % and Natural gas 58 % (IRENA, 2014). The electricity is mainly converted from fossil fuels, as shown in Figure 15. According to IRENA the country has high renewable energy resources of wind, solar, hydro and geothermal power (IRENA, 2014).

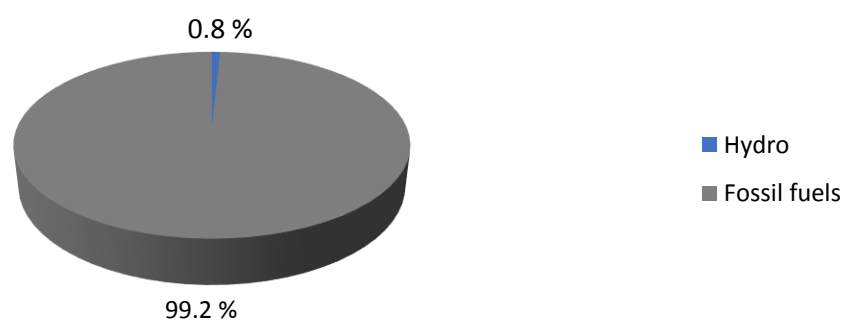


Figure 15: Electricity mix of Algeria 2011, data from (AfDB Group, 2013)

In Algeria it is the state utility Sonelgaz who owns the national grid and are responsible for production, transmission and distribution in the country. Algeria has interconnected their grid with Morocco, Tunisia and Libya and they have plans for submarine electricity cables with their European neighbours (Wuppertal Institute for climate, environment and energy, 2010).

## A.2 BURUNDI

|                                    |                                               |                                                                                                                                       |
|------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Official name: REPUBLIC OF BURUNDI |                                               |                                                   |
| Region:                            | Eastern Africa*                               |                                                                                                                                       |
| Government type:                   | Republic                                      |                                                                                                                                       |
| Currency:                          | Burundi francs (BIF)                          |                                                                                                                                       |
| Surface area, land:                | 27 830 km <sup>2</sup>                        |                                                                                                                                       |
| Population (2014 est.):            | 10 395 931                                    |                                                                                                                                       |
| GDP/capita (2013 est.):            | 5.75 billion USD**                            |                                                                                                                                       |
| Exports                            |                                               |                                                                                                                                       |
| - Total (2013 est.):               | 122.8 million USD                             |                                                                                                                                       |
| - Commodities:                     | Coffee, tea, sugar, cotton, hides             |                                                                                                                                       |
| Imports                            |                                               | Unspecified references (including map & flag):<br>(CIA, 2014)<br>*Geographical region by (UN Statistics Division, 2013)<br>**2013 USD |
| - Total (2013 est.):               | 867.2 million USD                             |                                                                                                                                       |
| - Commodities:                     | Capital goods, petroleum products, foodstuffs |                                                                                                                                       |

Burundi is a small country locating in east Central Africa below the equator. Burundi has a tropic climate with two rainy seasons (Mars- May and September- December) (CIA, 2014). The country is considered as a low income country and is dependent of financial support even though they have favourable natural resources. The extraction is limited partly because of the lack of electricity (World Bank, 2014a).

Year 2009 the total primary energy supply was 98.2 PJ, of which biomass represented 97 %, petroleum products 2 %, hydro 0.9 % and coal and coal products 0.1 %. The electricity is mainly converted from hydropower, which can be seen in Figure 16, the remaining is supplied by diesel engines. All the existing hydropower plants in Burundi are run-of-river except for Rwegura (World Bank, 2014b). The country has high renewable energy resources of solar and hydro power (IRENA, 2014).

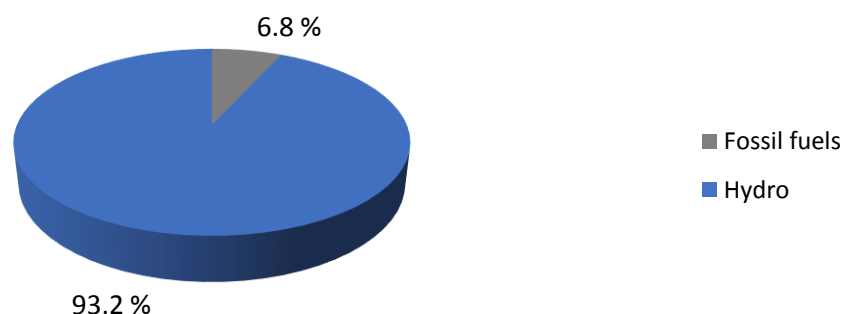


Figure 16: Electricity mix of Burundi 2011, data from (AfDB Group, 2013).

Régie de Production et de Distribution d'Eau et d'Electricité (REGIDESO) are a public owned company responsible for 94 % of the electricity production and distribution in Burundi. Beyond this, Burundi also buy 3.5 MW from Société Nationale d'Electricité (SNEL) who is the national power utility in Congo (DRC). Rural Electrification Agency of Burundi (ABER) is operating seven mini-plants which are not connected to the national grid (World Bank, 2014a).

### A.3 COMOROS

|                                     |                                                                                            |                                                                                                                                     |
|-------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Official name: UNION OF THE COMOROS |                                                                                            |                                                                                                                                     |
| Region:                             | Eastern Africa*                                                                            |                                                   |
| Government type:                    | Republic                                                                                   |                                                                                                                                     |
| Currency:                           | Comoran francs (KMF)                                                                       |                                                                                                                                     |
| Surface area, land:                 | 2 235 km <sup>2</sup>                                                                      |                                                                                                                                     |
| Population (2014 est.):             | 766 865                                                                                    |                                                                                                                                     |
| GDP/capita (2013 est.):             | 1 300 USD**                                                                                |                                                                                                                                     |
| Exports                             |                                                                                            |                                                                                                                                     |
| - Total (2013 est.):                | 19.7 million USD                                                                           |                                                                                                                                     |
| - Commodities:                      | Vanilla, ylang-ylang (perfume essence), cloves, copra                                      |                                                                                                                                     |
| Imports                             |                                                                                            |                                                                                                                                     |
| - Total (2013 est.):                | 208.8 million USD                                                                          | Unspecified references (including map): (CIA, 2014)<br>*Geographical region by (UN Statistics Division, 2013)<br>**Data in 2013 USD |
| - Commodities:                      | Rice and other foodstuffs, consumer goods, petroleum products, cement, transport equipment |                                                                                                                                     |

Comoros is a group of islands located in the Indian Ocean. The country gained their independence from France 1975. Comoros is considered to be a low income country (World Bank, 2014a) and is one of the world's poorest countries with a young population and few natural resources (CIA, 2014). However, the Comorian economy grew about 2.7 % in 2012 despite the ongoing euro crisis (AfDB Group, OECD dev & UNDP, 2013). The climate is tropical marine with a rainy season from November to May.

The majority of the energy used in Comoros is supplied by biomass. The total primary energy supply was 2008 about 4.5 PJ, of which biomass represented 58 %, petroleum products 41 % and hydro 0.4 % (IRENA, 2014).

The electricity is mainly converted from fossil fuels, as shown in Figure 17. According to IRENA the country has high renewable energy resources of geothermal, hydro and solar power (IRENA, 2014).

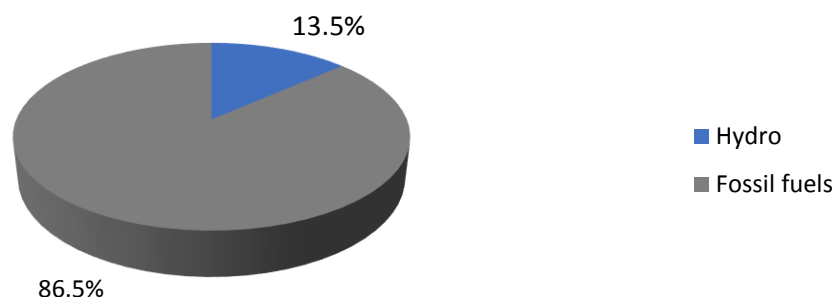


Figure 17: Electricity mix of Comoros 2011, data from (AfDB Group, 2013).

There are three grids in Comoros one on each island. The companies that are responsible for the electricity supply are MA-MWE who provides electricity to Moheli and Grande Comore and EDA which serves the Anjouan Island. For the moment there are no interconnections between the islands. The available capacity is 8 MW in Grande Comore, 1 MW in Moheli and 3-4 MW in EDA. The access rates of the islands are 60 % in Grand Comore, 50 % in Anjouan and 20 % in Moheli (World Bank, 2013b).

#### A.4 DJIBOUTI

|                                     |                                                                      |                                                                                                                                    |
|-------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Official name: REPUBLIC OF DJIBOUTI |                                                                      |                                                                                                                                    |
| Region:                             | Eastern Africa*                                                      |                                                                                                                                    |
| Government type:                    | Republic                                                             |                                                                                                                                    |
| Currency:                           | Djibouti francs (DJF)                                                |                                                                                                                                    |
| Surface area, land:                 | 23 200 km <sup>2</sup>                                               |                                                                                                                                    |
| Population (2014 est.):             | 810 179                                                              |                                                                                                                                    |
| GDP/capita (2013 est.):             | 1.459 billion USD**                                                  |                                                                                                                                    |
| Exports                             |                                                                      |                                                                                                                                    |
| - Total (2013 est.):                | 90.8 million USD                                                     |                                                                                                                                    |
| - Commodities:                      | Reexports, hides and skins, coffee                                   |                                                                                                                                    |
| Imports                             |                                                                      |                                                                                                                                    |
| - Total (2013 est.):                | 593.3 million USD                                                    | Unspecified references (including map & flag): (CIA, 2014)<br>*Geographical region by (UN Statistics Division, 2013)<br>**2013 USD |
| - Commodities:                      | Foods, beverages, transport equipment, chemicals, petroleum products |                                                                                                                                    |

Djibouti is a small country locating in northeast Africa against the Red Seas southern inlet. The country consist more less of deserts, volcanic rocks and salt lakes, this makes it difficult to grow things in Djibouti. The climate is dry and hot except in the high mountains, it rains hardly anything in this part of the world (CIA, 2014). Djibouti's economy is relayed on the service sector mainly from the harbour. The country is considered as a low income country (World Bank, 2014a).

Year 2009 the total primary energy supply was 11.2 PJ, of which biomass represented 32 % and petroleum products represent the remaining 68 %. The electricity is only converted from fossil fuels, which can be seen in Figure 18. The country has high renewable energy resources of solar and geothermal power (IRENA, 2014).

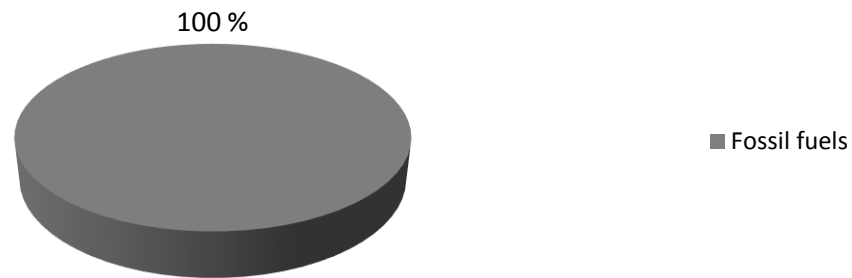


Figure 18: Electricity mix of Djibouti 2011, data from (AfDB Group, 2013).

In Djibouti there is a state-owned company, Electricité de Djibouti (EDD), that has the monopoly on generation, distribution and marketing of the electricity in most of the country. EDD is unable to satisfy all domestic demand why there are also some private firms (Electricité de Djibouti, 2014).

## A.5 LIBYA

|                         |                                                                               |                                                                                                                                    |
|-------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Official name:          |                                                                               | <b>LIBYA</b>                                                                                                                       |
| Region:                 | Northern Africa*                                                              |                                                  |
| Government type:        | Operated under a transitional government                                      |                                                                                                                                    |
| Currency:               | Libyan dinars (LYD)                                                           |                                                                                                                                    |
| Surface area, land:     | 1 759 540 km <sup>2</sup>                                                     |                                                                                                                                    |
| Population (2014 est.): | 6 244 174                                                                     |                                                                                                                                    |
| GDP/capita (2013 est.): | 70.92 billion USD**                                                           |                                                                                                                                    |
| Exports                 |                                                                               |                                                                                                                                    |
| - Total (2013 est.):    | 38.45 billion USD                                                             |                                                                                                                                    |
| - Commodities:          | Crude oil, refined petroleum products, natural gas, chemicals.                |                                                                                                                                    |
| Imports                 |                                                                               |                                                                                                                                    |
| - Total (2013 est.):    | 27.15 billion USD                                                             | Unspecified references (including map & flag): (CIA, 2014)<br>*Geographical region by (UN Statistics Division, 2013)<br>**2013 USD |
| - Commodities:          | Machinery, semi-finished goods, food, transport equipment, consumer products. |                                                                                                                                    |

Libya is located in Northern Africa with a coast line towards the Mediterranean Sea and consists of 90 % desert. Libya has desert climate inland and Mediterranean climate along the coast (CIA, 2014). Libya's economy totally rely on exports of natural resources and they are today considered as an upper middle income country (World Bank, 2014a).

Year 2009 the total primary energy supply was 854.3 PJ, of which oil and oil products represented 72 %, natural gas 27 % and biomass 1 %. The electricity is generated from fossil fuels as can be seen in Figure 19. The country has high renewable energy resources of solar and wind power (IRENA, 2014).



Figure 19: Electricity mix of Libya 2011, data from (AfDB Group, 2013).

General Electricity Company of Libya (GECOL) is the electric utility who is responsible for production, transmission and distribution in the whole region. They have more than 30 power plants running continuously (mainly steam turbines, simple cycle gas turbines and generators) (ABB Switzerland Ltd, 2009).

## A.6 MALI

| Official name: <b>REPUBLIC OF MALI</b> |                                                                                  |                                                            |
|----------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Region:                                | Western Africa*                                                                  |                                                                                                                                               |
| Government type:                       | Republic                                                                         |                                                                                                                                               |
| Currency:                              | Burundi francs (BIF)                                                             |                                                                                                                                               |
| Surface area, land:                    | 1 240 192 km <sup>2</sup>                                                        |                                                                                                                                               |
| Population (2014 est.):                | 16 455 903                                                                       |                                                                                                                                               |
| GDP/capita (2013 est.):                | 11.37 billion USD**                                                              |                                                                                                                                               |
| Exports                                |                                                                                  |                                                                                                                                               |
| - Total (2013 est.):                   | 2.577 billion USD                                                                |                                                                                                                                               |
| - Commodities:                         | Cotton, gold, livestock                                                          |                                                                                                                                               |
| Imports                                |                                                                                  | Unspecified references (including map & flag):<br>(CIA, 2014)<br><br>*Geographical region by (UN Statistics Division, 2013)<br><br>**2013 USD |
| - Total (2013 est.):                   | 2.895 billion USD                                                                |                                                                                                                                               |
| - Commodities:                         | Petroleum, machinery and equipment, contraction materials, foodstuffs, textiles. |                                                                                                                                               |

Mali is located in Western Africa and consists primarily of desert and semi-desert. The northern Mali is part of the Sahara desert, and in the southern parts there are the rivers Niger and Senegal and a rainforest. The climate is hot and dry but it is somewhat wetter in the southern parts of the country. They have two seasons one extremely dry and one rainy season (CIA, 2014). Mali is considered as a low income country and is dependent of financial support (World Bank, 2014a).

Year 2008 the total primary energy supply was 135.0 PJ, of which biomass represented 78 %, oil and oil products was 21% and hydro 1 %. The electricity is mainly converted from fossil fuel, which can be seen in Figure 20. The country has high renewable energy resources of wind, solar and hydro power (IRENA, 2014).

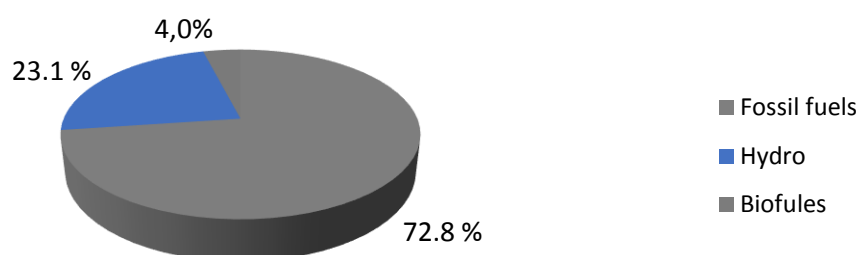


Figure 20: Electricity mix of Mali 2011, data from (AfDB Group, 2013).

The electrification in Mali is low, most households in the rural areas are still satisfied their energy need with kerosene and batteries. There is a national grid regulated by Engenie du Mali SA (EDM) together with 21 local energy companies for isolated centres. Malian grid is today connected to both Mauritania and Senegal (Republic of Mali- Ministry of energy and water, 2011).

## A.7 MOROCCO

|                         |                                                                                                                                                      |  |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Official name:          | <b>KINGDOM OF MOROCCO</b>                                                                                                                            |  |
| Region:                 | Northern Africa*                                                                                                                                     |  |
| Government type:        | Constitutional monarchy                                                                                                                              |  |
| Currency:               | Moroccan dirhams (MAD)                                                                                                                               |  |
| Surface area, land:     | 446 550 km <sup>2</sup>                                                                                                                              |  |
| Population (2014 est.): | 32 987 206                                                                                                                                           |  |
| GDP/capita (2013 est.): | 104.8 billion USD**                                                                                                                                  |  |
| Exports                 |                                                                                                                                                      |  |
| - Total (2013 est.):    | 16.78 billion USD                                                                                                                                    |  |
| - Commodities:          | Clothing and textiles, electric components, inorganic chemicals, transistors, crude minerals, petroleum products, citrus fruits, vegetables and fish |  |
| Imports                 |                                                                                                                                                      |  |
| - Total (2013 est.):    | 38.88 billion USD                                                                                                                                    |  |
| - Commodities:          | Crude petroleum, textile fabric, telecommunications equipment, wheat, gas, electricity, transistors and plastics.                                    |  |



Unspecified references (including map & flag): (CIA, 2014)

\*Geographical region by (UN Statistics Division, 2013)

\*\*2013 USD

Morocco is located in Northern Africa and is bordered to east by Algeria and Western Sahara in the south. Morocco has a hot and dry climate but the temperature can differ quite much between valleys and mountain peaks, there is both snow on the mountains and mess spirit desert heat (CIA, 2014). Mali is considered as a low middle income country and is dependent of financial support (World Bank, 2014a).

Year 2009 the total primary energy supply was 631.5 PJ, of which oil and oil products represented 73 %, coal and coal product 18 %, natural gas 4%, biomass 3 % and hydro 2 %. The electricity is converted from fossil fuel, hydro and other renewable as can be seen in Figure 21. The country has high renewable energy resources of wind, solar and hydro power (IRENA, 2014).

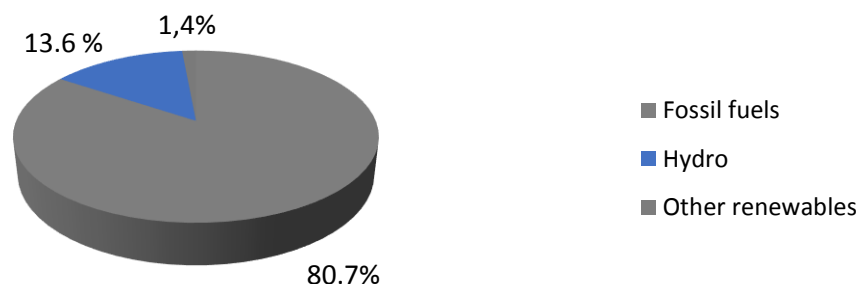


Figure 21: Electricity mix of Morocco 2011, data from (AfDB Group, 2013).

It is the public utility L'Office National de l'Electricité (ONE) how is responsible for production, transmission and distribution in Morocco (Paving the way for MSP, 2014).

## A.8 SAO TOME AND PRINCIPE

|                         |                                                                       |                                                                                                                                      |
|-------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Official name:          | <b>DEMOCRATIC REPUBLIC OF SAO TOME AND PRINCIPE</b>                   |                                                                                                                                      |
| Region:                 | Middle Africa*                                                        |                                                    |
| Government type:        | Republic                                                              |                                                                                                                                      |
| Currency:               | Dobras (STD)                                                          |                                                                                                                                      |
| Surface area, land:     | 964 km <sup>2</sup>                                                   |                                                                                                                                      |
| Population (2014 est.): | 190 428                                                               |                                                                                                                                      |
| GDP/capita (2013 est.): | 2 200 USD**                                                           |                                                                                                                                      |
| Exports                 |                                                                       |                                                                                                                                      |
| - Total (2013 est.):    | 12.1 million USD                                                      |                                                                                                                                      |
| - Commodities:          | Cocoa (80 %), copra coffee, palm oil                                  |                                                                                                                                      |
| Imports                 |                                                                       |                                                                                                                                      |
| - Total (2013 est.):    | 112.8 million USD                                                     | Unspecified references (including map): (CIA, 2014).<br>*Geographical region by (UN Statistics Division, 2013)<br>**Data in 2013 USD |
| - Commodities:          | Machinery and electrical equipment, food products, petroleum products |                                                                                                                                      |

The country Sao Tome and Principe, named after its two major islands, is located in the Gulf of Guinea just over the equator. The islands have a tropical climate with a rainy season between Octobers and May (CIA, 2014). Sao Tome and Principe is a low middle income country (World Bank, 2014a) which has environmental issues as deforestation, soil erosion and exhaustion. The economy of the country is since its independence 1975 in a large extent dependent on cocoa.

The total primary energy supply was year 2008 3.0 PJ, of which petroleum products made 62 %, biomass 37 % and hydro 1 % (IRENA, 2014). As seen in Figure 22 the electricity is conversion is based on fossil fuels and hydro power. The potential of renewable energy sources is high for solar power (IRENA, 2014). There is also a potential for petroleum resources in the territorial waters of the country, but practical extraction is several years of (CIA, 2014).

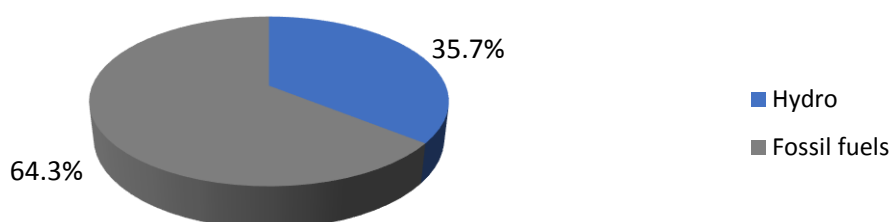


Figure 22: Electricity mix of Sao Tome and Principe 2011, data from (AfDB Group, 2013).

In Sao Tome and Principe the power plants are operated by the national electricity corporation EMAE (Empresa de Agua e Electricidade) (African Development Fund, 2004). Since the grid is not sufficiently reliable a large amount of the consumers use off-grid solutions (especially diesel generators) (UNIDO, 2013).

## A.9 SEYCHELLES

|                         |                                                                                              |                                                                                                                                     |  |
|-------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--|
| Official name:          |                                                                                              | <b>REPUBLIC OF SEYCHELLES</b>                                                                                                       |  |
| Region:                 | Eastern Africa*                                                                              |                                                  |  |
| Government type:        | Republic                                                                                     |                                                                                                                                     |  |
| Currency:               | Seychelles rupees (SCR)                                                                      |                                                                                                                                     |  |
| Surface area, land:     | 455 km <sup>2</sup>                                                                          |                                                                                                                                     |  |
| Population (2014 est.): | 91 650                                                                                       |                                                                                                                                     |  |
| GDP/capita (2013 est.): | 25 900 USD**                                                                                 |                                                                                                                                     |  |
| Exports                 |                                                                                              |                                                                                                                                     |  |
| - Total (2013 est.):    | 516.7 million USD                                                                            |                                                                                                                                     |  |
| - Commodities:          | Canned tuna, frozen fish, cinnamon bark, copra, petroleum products (reexports)               |                                                                                                                                     |  |
| Imports                 |                                                                                              |                                                                                                                                     |  |
| - Total (2013 est.):    | 845.4 million USD                                                                            |                                                                                                                                     |  |
| - Commodities:          | Machinery and equipment, foodstuffs, petroleum products, chemicals, other manufactured goods | Unspecified references (including map): (CIA, 2014)<br>*Geographical region by (UN Statistics Division, 2013)<br>**Data in 2013 USD |  |

Seychelles consist of 115 small islands (of which 30 of them are inhabited) in the Indian Ocean outside of Africa's east coast south of the equator. They have a tropic climate with high temperatures and humidity and they have two seasons depending on the monsoon winds. Their winter season is between May and October, during this season they have a dry weather. Their summer is between November to April with a lot of rain, especially during January to February (Government of Seychelles, 2011). Seychelles became independent in 1976 and is today classified as an upper middle income country and their economy relies on fishing and tourism (World Bank, 2014a).

The total primary energy supply in Seychelles was year 2009 10.1 PJ, of which petroleum products made 100 % (IRENA, 2014). As can be seen in Figure 23 the electricity production is also based on fossil fuels. The potential of renewable energy sources is high for solar power (IRENA, 2014).

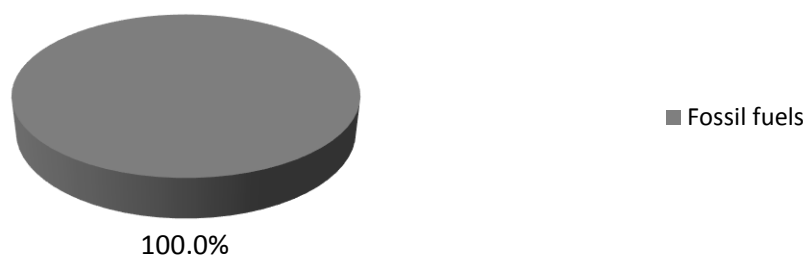


Figure 23: Electricity mix of Seychelles 2011, data from (AfDB Group, 2013).

The Three biggest islands, Mahe and Praslin have their own separated grid system. The Public Utility Company (PUC) is responsible for production and distribution of electricity in Mahe and Praslin. Mahe feed some of the inner islands via a sub-marine distribution cable and Praslin supplies La Digue via a sub-marine cable. Seychelles are relaying their electricity production on imported fossil fuels. The renewable electricity alternatives are limited to solar power, water heaters for bathing, and some biomass (wood) for cooking (Government of Seychelles, 2011).

## A.10 SOUTH SUDAN

|                         |                                |                                                                                                                                      |
|-------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Official name:          | <b>REPUBLIC OF SOUTH SUDAN</b> |                                                                                                                                      |
| Region:                 | Eastern Africa*                |                                                    |
| Government type:        | Republic                       |                                                                                                                                      |
| Currency:               | South Sudanese pounds (SSP)    |                                                                                                                                      |
| Surface area, land:     | 644 329 km <sup>2</sup>        |                                                                                                                                      |
| Population (2014 est.): | 11 562 695                     |                                                                                                                                      |
| GDP/capita (2013 est.): | 1 400 USD**                    |                                                                                                                                      |
| Exports                 |                                |                                                                                                                                      |
| - Total (2013 est.):    | No data                        |                                                                                                                                      |
| - Commodities:          | Oil                            |                                                                                                                                      |
| Imports                 |                                |                                                                                                                                      |
| - Total (2013 est.):    | No data                        |                                                                                                                                      |
| - Commodities:          | Goods, services, capital       | Unspecified references (including map): (CIA, 2014)<br>*Geographical regions by (UN Statistics Division, 2013)<br>**Data in 2013 USD |

South Sudan is a young country which independence from Sudan was attained in July 2011 after several decades of civil war (CIA, 2014). The country is considered to be a low income country (World Bank, 2014a). The climate of South Sudan is hot with seasonal rainfall. The Sudd (see map above) covers about 15 % of the country areas and is one of the largest wetlands in the world (CIA, 2014). The economy has been going down since 2012 when the government decided to shut down oil production due to disputes with Sudan, and the country are fighting several problems related to the weak economy.

Data about the total primary energy supply 2008 can be found in the following introductory text about Sudan, since South Sudan at that time was included in Sudan. Less than 1 % of the population does today have access to electricity (Minister of transport, electricity and dams, housing and physical planning, 2012).

As seen in Figure 24, the majority of the electricity generated in South Sudan today is from fossil fuels with a small share of renewable energy, probably mainly from solar power. However, there are several hydro power plants in South Sudan, 2010 hydroelectric plants made about 66 % of the total installed electricity capacity (World Bank, 2014a). For information about renewable energy sources, please read the introductory text about Sudan below.

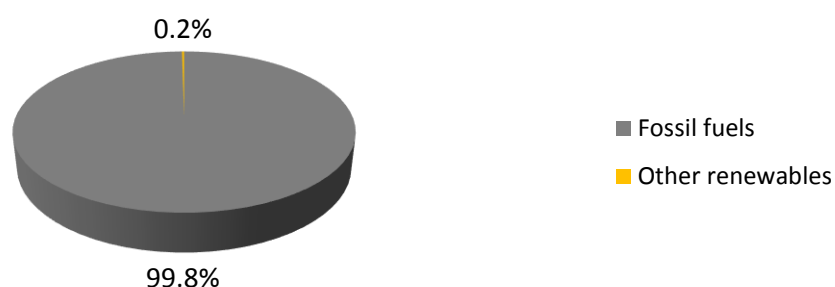


Figure 24: Electricity mix of South Sudan 2011, data from (AfDB Group, 2013).

### A.11 SUDAN

|                         |                                                                                                            |                                                                                                                                      |                                                                                    |
|-------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Official name:          |                                                                                                            |                                                                                                                                      | REPUBLIC OF THE SUDAN                                                              |
| Region:                 | Northern Africa*                                                                                           |                                                                                                                                      |  |
| Government type:        | Federal republic                                                                                           |                                                                                                                                      |                                                                                    |
| Currency:               | Sudanese pounds (SDG)                                                                                      |                                                                                                                                      |                                                                                    |
| Surface area, land:     | 1 861 484 km <sup>2</sup>                                                                                  |                                                                                                                                      |                                                                                    |
| Population (2014 est.): | 35 482 233                                                                                                 |                                                                                                                                      |                                                                                    |
| GDP/capita (2013 est.): | 2 600 USD*                                                                                                 |                                                                                                                                      |                                                                                    |
| Exports                 |                                                                                                            |                                                                                                                                      |                                                                                    |
| - Total (2013 est.):    | 4.145 billion USD                                                                                          |                                                                                                                                      |                                                                                    |
| - Commodities:          | Gold, oil and petroleum product, cotton, sesame, livestock, groundnuts, gum arabic, sugar                  |                                                                                                                                      |                                                                                    |
| Imports                 |                                                                                                            |                                                                                                                                      |                                                                                    |
| - Total (2013 est.):    | 5.941 billion USD                                                                                          |                                                                                                                                      |                                                                                    |
| - Commodities:          | Foodstuffs, manufactured goods, refinery and transport equipment, medicines and chemicals, textiles, wheat | Unspecified references (including map): (CIA, 2014)<br>*Geographical regions by (UN Statistics Division, 2013)<br>**Data in 2013 USD |                                                                                    |

Sudan is located in northern east Africa with a short coastline, on the eastern side of the country, towards the Red Sea. The country is dominated by the Sahara Desert, but in the south there is some shrub savanna. The climate is really hot and dry, especially in the northern parts (around the deserts) but it can be cold under nights during the winter season (East African Power Pool (EAPP) and East African Community (EAC), 2011).

Sudan counts as a lower middle income country (World Bank, 2014a). The total primary energy supply was, 2009, 662.2 PJ of which biomass made 68 %, hydro 2 % and petroleum products made 30 % (IRENA, 2014). As seen in Figure 25 the electricity is conversion is based on fossil fuels, hydro power and Biofuels. The potential of renewable energy sources is high for solar, wind, hydro and geothermal power (IRENA, 2014).

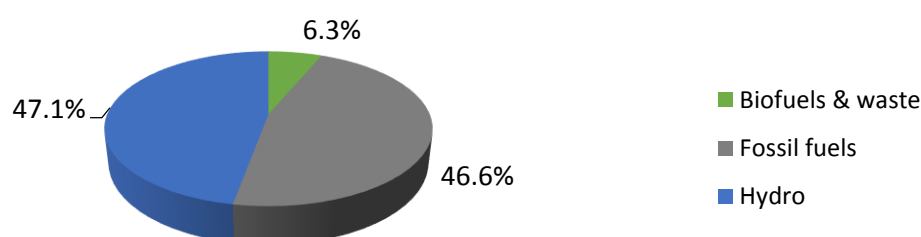


Figure 25: Electricity mix of Sudan 2011, data from (AfDB Group, 2013).

The National Electricity Corporation (NEC) is responsible for generation, transmission and distribution in Sudan. There are two grids in Sudan, the Blue Nile Grid and the Western grid which encompasses a small part of the country. The part of the country that is not covered by the national grid is relying on diesel-fired generators (East African Power Pool (EAPP) and East African Community (EAC), 2011).

## A.12 SWAZILAND

|                         |                                                                                                |                                                                                     |
|-------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Official name:          |                                                                                                | <b>KINGDOM SWAZILAND</b>                                                            |
| Region:                 | Sothorn Africa*                                                                                |  |
| Government type:        | Monarchy                                                                                       |                                                                                     |
| Currency:               | Emalangeni                                                                                     |                                                                                     |
| Surface area, land:     | 17 364 km <sup>2</sup>                                                                         |                                                                                     |
| Population (2014 est.): | 1 419 623                                                                                      |                                                                                     |
| GDP/capita (2013 est.): | 3.807 billion USD**                                                                            |                                                                                     |
| Exports                 |                                                                                                |                                                                                     |
| - Total (2013 est.):    | 1.603 billion USD                                                                              |                                                                                     |
| - Commodities:          | Soft drink concentrates, sugar, wood pulp, cotton yard, refrigerators, citrus and canned fruit |                                                                                     |
| Imports                 |                                                                                                |                                                                                     |
| - Total (2013 est.):    | 1.545 billion USD                                                                              | Unspecified references (including map & flag): (CIA, 2014)                          |
| - Commodities:          | Motor vehicles, machinery, transport equipment, foodstuffs, petroleum products and chemicals   | *Geographical region by (UN Statistics Division, 2013)<br>**2013 USD                |

Swaziland is located in Southern Africa and is bordered by Mozambique and South Africa. There is a hot subtropical climate here (CIA, 2014). Swaziland has natural resources such as coal, diamonds and gold. This country is considered a low middle income country and is dependent of financial support (World Bank, 2014a). Year 2009 the total primary energy supply was 94.0 PJ, of which biomass represented 63 %, oil and oil products 9 %, coal and coal product 5 % and heat 23%. The electricity is converted from fossil fuel, hydro and biofuels as can be seen in Figure 26. The country has high renewable energy resources of solar power (IRENA, 2014).

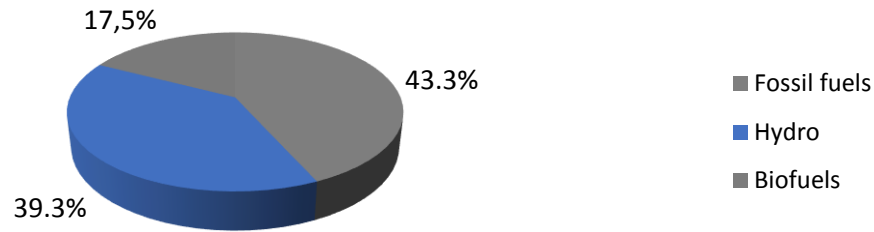


Figure 26: Electricity mix of Swaziland 2011, data from (AfDB Group, 2013).

Swaziland is importing most of their electricity from South Africa, around 40 % on the demand is produced locally by hydro power and biomass is most used outside the national grid. Swaziland Electricity Board (MNRE) is responsible for producing, transmission and distribution in the country (Commission of European Communities, 2006).

### A.13 TOGO

| Official name: <b>TOGOLESE REPUBLIC</b> |                                                         |                                                                                      |
|-----------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------|
| Region:                                 | Western Africa*                                         |   |
| Government type:                        | Republic                                                |                                                                                      |
| Currency:                               | Communaute Financiere Africaine francs (XOF)            |                                                                                      |
| Surface area, land:                     | 56 785 km <sup>2</sup>                                  |                                                                                      |
| Population (2014 est.):                 | 7 351 374                                               |                                                                                      |
| GDP/capita (2013 est.):                 | 4.299 billion USD**                                     |                                                                                      |
| Exports                                 |                                                         |                                                                                      |
| - Total (2013 est.):                    | 982.2 million USD                                       |                                                                                      |
| - Commodities:                          | Reexports, cotton, phosphates, coffee, cocoa            |                                                                                      |
| Imports                                 |                                                         |                                                                                      |
| - Total (2013 est.):                    | 1.677 million USD                                       |  |
| - Commodities:                          | Machinery and equipment, foodstuffs, petroleum products |                                                                                      |

Unspecified references (including map & flag): (CIA, 2014)  
 \*Geographical region by (UN Statistics Division, 2013)  
 \*\*2013 USD

Togo is located in Western Africa, bordering Benin, Ghana, and Burkina Faso. The climate in Togo is tropical; the northern part of the country has one rainy season (between June and September) and the southern part has two rainy seasons, one in April to June and the other in September to October. In December and January, desert winds prevail (CIA, 2014). Togo has natural resources like phosphate and is considered a low-income country (World Bank, 2014a).

In 2009, the total primary energy supply was 110.0 PJ, of which biomass represented 85 %, oil and oil products 14.7 % and hydro 0.3 %. The electricity is mainly generated from hydropower, which can be seen in Figure 27; the remaining is supplied by fossil fuels and biofuels. The country has a high renewable energy resource of hydro power (IRENA, 2014).

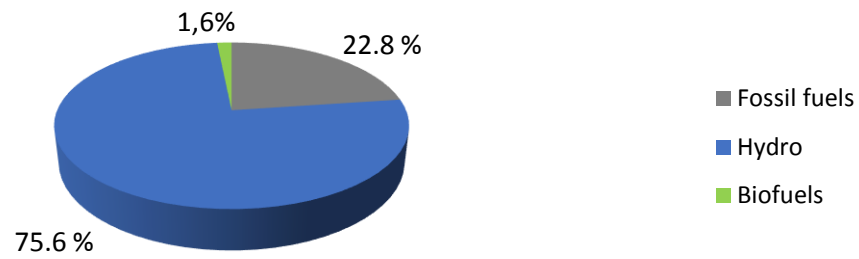


Figure 27: Electricity mix of Togo 2011, data from (AfDB Group, 2013).

There are two companies in Togo that are responsible for the electricity in the country, Compagnie Energie Electrique di Togo (CEET) and Communauté du Bénin (CEB) (PPIAF, 2011).

## A.14 TUNISIA

|                         |                                                                                                                                                     |                                                                                                                                    |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Official name:          |                                                                                                                                                     | <b>TUNISIA REPUBLIC</b>                                                                                                            |
| Region:                 | Northern Africa*                                                                                                                                    |                                                 |
| Government type:        | Republic                                                                                                                                            |                                                                                                                                    |
| Currency:               | Tunisian dinars (TND)                                                                                                                               |                                                                                                                                    |
| Surface area, land:     | 163 610 km <sup>2</sup>                                                                                                                             |                                                                                                                                    |
| Population (2014 est.): | 10 937 521                                                                                                                                          |                                                                                                                                    |
| GDP/capita (2013 est.): | 48.38 billion USD**                                                                                                                                 |                                                                                                                                    |
| Exports                 |                                                                                                                                                     |                                                                                                                                    |
| - Total (2013 est.):    | 17.46 billion USD                                                                                                                                   |                                                                                                                                    |
| - Commodities:          | Clothing, semi-finished goods and textiles, agricultural products, mechanical goods, phosphate and chemicals, hydrocarbons and electrical equipment |                                                                                                                                    |
| Imports                 |                                                                                                                                                     |                                                                                                                                    |
| - Total (2013 est.):    | 24.95 billion USD                                                                                                                                   | Unspecified references (including map & flag): (CIA, 2014)<br>*Geographical region by (UN Statistics Division, 2013)<br>**2013 USD |
| - Commodities:          | Textiles, machinery and equipment, hydrocarbons, chemicals and foodstuffs.                                                                          |                                                                                                                                    |

Tunisia is located in Northern Africa bordering to Algeria and Libya, they also have coastline towards the Mediterranean Sea. Northern Tunisia has a tropical climate with hot and dry summers and relatively rainy winters further south is the climate more dry and the most southern regions have a desert climate (CIA, 2014). Tunisia's natural resources consist of phosphate, oil and natural gas. Sadly, they are not self-sufficient and are therefore importing a lot of oil and natural gas for their energy production. Tunisia is considered as an upper middle income country (World Bank, 2014a). Year 2009 the total primary energy supply was 385.2 PJ, of which natural gas represented 46 %, oil and oil products 40 %, biomass 13.8 %, wind 0.1 % and hydro 0.1 %. The electricity is mainly converted from fossil fuels, which can be seen in Figure 28. The country has high renewable energy resources of wind, solar and hydro power (IRENA, 2014).

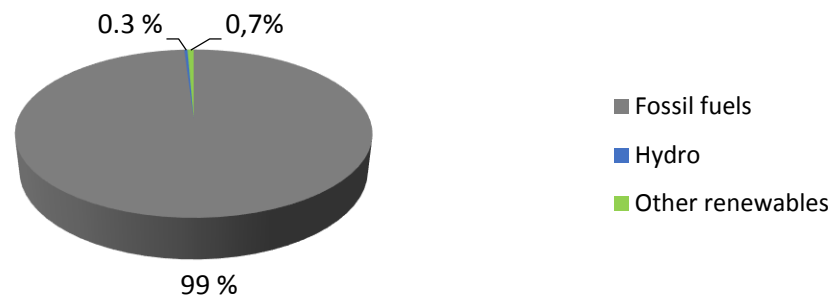


Figure 28: Electricity mix of Tunisia 2011, data from (AfDB Group, 2013).

In Tunisia it is STEG (Tunisian Company for Electricity and Gas) who is responsible for the production, transmission and distribution of the electricity. Tunisia has an interconnection with Algerian they are also working on two links with Libya (Bernedetti et al., 2013).

## APPENDIX B - DEFAULT DATA

Data present in this chapter have been used for all the simulated countries. References can be found in *Appendix D – References for appendix B & C*.

### B.1 Fuel price projections

Table 2: Imported fuel prices

| Costs<br>[MUSD/PJ] | Coal | Diesel | HFO   | Natural Gas | Biogas/landfill gas | Biomass | Uranium | Imported/Exported<br>electricity |
|--------------------|------|--------|-------|-------------|---------------------|---------|---------|----------------------------------|
| 2010               | 3.96 | 15.3   | 14.96 | 6.38        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2011               | 3.96 | 15.3   | 14.96 | 6.38        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2012               | 3.96 | 15.3   | 14.96 | 6.38        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2013               | 3.96 | 15.3   | 14.96 | 6.38        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2014               | 3.63 | 16     | 15.73 | 6.38        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2015               | 3.74 | 16.4   | 16.17 | 6.49        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2016               | 3.63 | 17.2   | 16.61 | 6.6         | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2017               | 3.74 | 17.8   | 17.16 | 6.71        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2018               | 3.63 | 18.4   | 17.6  | 6.71        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2019               | 3.63 | 18.8   | 17.82 | 6.82        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2020               | 3.63 | 19.1   | 17.93 | 6.93        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2021               | 3.63 | 19.3   | 18.15 | 7.04        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2022               | 3.63 | 19.4   | 18.37 | 7.26        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2023               | 3.63 | 19.7   | 18.59 | 7.26        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2024               | 3.63 | 19.9   | 18.81 | 7.26        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2025               | 3.63 | 20.2   | 19.03 | 7.26        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2026               | 3.63 | 20.4   | 19.14 | 7.48        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2027               | 3.63 | 20.6   | 19.47 | 7.59        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2028               | 3.63 | 20.9   | 19.69 | 7.81        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2029               | 3.63 | 21.3   | 20.02 | 8.14        | 15.28               | 5.1     | 2.5     | 22.22                            |
| 2030               | 3.74 | 21.5   | 20.35 | 8.36        | 15.28               | 5.1     | 2.5     | 22.22                            |
| Reference:         | 1    | 1      | 1     | 1           | 2                   | 1       | 4       | 1                                |

Table 3: Fuel prices local resources/extraction

| Costs<br>[MUSD/PJ] | Coal | Diesel | HFO  | Natural Gas | Biogas/landfill gas | Biomass | Uranium | Imported/Exported<br>electricity |
|--------------------|------|--------|------|-------------|---------------------|---------|---------|----------------------------------|
| 2010               | 3.6  | 16.83  | 13.6 | 5.8         | 13.89               | 2.27    | 1.75    | -                                |
| 2011               | 3.6  | 16.83  | 13.6 | 5.8         | 13.89               | 2.27    | 1.75    | -                                |
| 2012               | 3.6  | 16.83  | 13.6 | 5.8         | 13.89               | 2.27    | 1.75    | -                                |
| 2013               | 3.6  | 16.83  | 13.6 | 5.8         | 13.89               | 2.27    | 1.75    | -                                |
| 2014               | 3.3  | 17.6   | 14.3 | 5.8         | 13.89               | 2.27    | 1.75    | -                                |
| 2015               | 3.4  | 18.04  | 14.7 | 5.9         | 13.89               | 2.27    | 1.75    | -                                |
| 2016               | 3.3  | 18.92  | 15.1 | 6           | 13.89               | 2.27    | 1.75    | -                                |
| 2017               | 3.4  | 19.58  | 15.6 | 6.1         | 13.89               | 2.27    | 1.75    | -                                |
| 2018               | 3.3  | 20.24  | 16   | 6.1         | 13.89               | 2.27    | 1.75    | -                                |
| 2019               | 3.3  | 20.68  | 16.2 | 6.2         | 13.89               | 2.27    | 1.75    | -                                |
| 2020               | 3.3  | 21.01  | 16.3 | 6.3         | 13.89               | 2.27    | 1.75    | -                                |
| 2021               | 3.3  | 21.23  | 16.5 | 6.4         | 13.89               | 2.27    | 1.75    | -                                |
| 2022               | 3.3  | 21.34  | 16.7 | 6.6         | 13.89               | 2.27    | 1.75    | -                                |
| 2023               | 3.3  | 21.67  | 16.9 | 6.6         | 13.89               | 2.27    | 1.75    | -                                |
| 2024               | 3.3  | 21.89  | 17.1 | 6.6         | 13.89               | 2.27    | 1.75    | -                                |
| 2025               | 3.3  | 22.22  | 17.3 | 6.6         | 13.89               | 2.27    | 1.75    | -                                |
| 2026               | 3.3  | 22.44  | 17.4 | 6.8         | 13.89               | 2.27    | 1.75    | -                                |
| 2027               | 3.3  | 22.66  | 17.7 | 6.9         | 13.89               | 2.27    | 1.75    | -                                |
| 2028               | 3.3  | 22.99  | 17.9 | 7.1         | 13.89               | 2.27    | 1.75    | -                                |
| 2029               | 3.3  | 23.43  | 18.2 | 7.4         | 13.89               | 2.27    | 1.75    | -                                |
| 2030               | 3.4  | 23.65  | 18.5 | 7.6         | 13.89               | 2.27    | 1.75    | -                                |
| Reference:         | 1    | 1      | 1    | 1           | 2                   | 1       | 4       | -                                |

## B.2 Technology data

Table 4: Transmission and distribution data used as a standard for all countries

| Technology data                  | Distribution to Industry | Distribution to Rural Areas | Distribution to Urban Areas | Transmission |
|----------------------------------|--------------------------|-----------------------------|-----------------------------|--------------|
| Parameter                        |                          |                             |                             |              |
| Power plant lifetime [years]     | 60                       | 60                          | 60                          | 60           |
| Capacity factor/plant factor [%] | 100                      | 100                         | 100                         | 100          |
| Availability factor [%]          | 100                      | 100                         | 100                         | 100          |
| Capital Cost [MUSD/GW]           | 840.4                    | 4233.60                     | 2433.30                     | 365          |
| Variable O&M [MUSD/TWh]          | 0                        | 0                           | 0                           | 0            |
| Fixed O&M [MUSD/GW]              | 0                        | 0                           | 0                           | 0            |
| Reference:                       | 6                        | 6                           | 6                           | 6            |

Table 5: Technology data for biogas, biomass/coal, biomass, and coal and HFO power plant

| Technology data              | Biogas  | Biomass/coal | Biomass | Coal    | HFO OCGT |
|------------------------------|---------|--------------|---------|---------|----------|
| Parameter                    |         |              |         |         |          |
| Efficiency [%]               | 32      | 38.5         | 38      | 37      | 35       |
| Power plant lifetime [years] | 25      | 30           | 30      | 35      | 25       |
| Capacity-/plant factor [%]   | 77.5    | 70           | 50      | 85      | 90       |
| Availability factor [%]      | 80      | 90           | 93      | 94      | 90       |
| Capital Cost [MUSD/GW]       | 5514.62 | 592.90       | 3660.25 | 3518.79 | 1633.50  |
| Variable O&M [MUSD/TWh]      | 1.20    | 0            | 5.56    | 3.96    | 4.17     |
| Fixed O&M [MUSD/GW]          | 303.84  | 17.787       | 17.79   | 0       | 0        |
| Reference:                   | 8       | 8            | 7       | 7       | 7        |

Table 6: Technology data for different diesel power plants and natural gas power plants.

| Technology data              | Diesel DE | Diesel DE,<br>Decentralized industry | Diesel DE,<br>Decentralized Rural areas | Diesel DE,<br>Decentralized Urban areas | NG CCGT | NG OCGT |
|------------------------------|-----------|--------------------------------------|-----------------------------------------|-----------------------------------------|---------|---------|
| Parameter                    |           |                                      |                                         |                                         |         |         |
| Efficiency [%]               | 35        | 35                                   | 16                                      | 16                                      | 48      | 30      |
| Power plant lifetime [years] | 25        | 20                                   | 10                                      | 10                                      | 30      | 25      |
| Capacity-/plant factor [%]   | 90        | 90                                   | 90                                      | 90                                      | 93.50   | 93.50   |
| Availability factor [%]      | 90        | 90                                   | 90                                      | 90                                      | 93      | 93      |
| Capital Cost [MUSD/GW]       | 1177.00   | 658.70                               | 691.88                                  | 691.88                                  | 1422.84 | 729.63  |
| Variable O&M [MUSD/TWh]      | 4.72      | 15.37                                | 9.23                                    | 9.23                                    | 0.804   | 5.534   |
| Fixed O&M [MUSD/GW]          | 0         | 0                                    | 0                                       | 0                                       | 0       | 0       |
| Reference:                   | 7         | 7                                    | 7                                       | 7                                       | 7       | 7       |

Table 7: Technology data for different sizes of hydro and wind power plant and for nuclear power plant.

| Technology data                                   | Hydro, Small | Hydro, medium | Hydro , large | Nuclear   | Wind , 30 % | Wind, 25 % |
|---------------------------------------------------|--------------|---------------|---------------|-----------|-------------|------------|
| Parameter                                         |              |               |               |           |             |            |
| Efficiency [%]                                    | 100          | 100           | 100           | 33        | 100         | 100        |
| Power plant lifetime [years]                      | 50           | 50            | 50            | 60        | 25          | 25         |
| Capacity-/plant factor [%]                        | 50           | 45            | 90            | 85        | 30          | 25         |
| Availability factor [%]                           | 98           | 98            | 90            | 100       | 85          | 90         |
| Capital Cost [MUSD/GW]<br>(2010, 2020 and 2030) * | 5396.14      | 5093.57       | 6370.79       | 10 777.96 | 2861.65     | 2607.41    |
|                                                   | 4856.52      | 4527.16       | 5733.71       | 10 777.96 | 2861.65     | 2607.41    |
|                                                   | 4316.91      | 4774.84       | 5733.71       | 10 777.96 | 2861.65     | 2607.41    |
| Variable O&M [MUSD/TWh]                           | 0            | 0             | 0             | 3.87012   | 3.96815     | 3.96810    |
| Fixed O&M [MUSD/GW]                               | 75           | 65            | 60            | 0         | 0           | 0          |
| Reference:                                        | 7            | 7             | 7             | 7         | 7           | 7          |

\*For Egypt hydro power plant specific data is available why the data above only is used for Madagascar, Mauritius and Zimbabwe.

Table 8: Technology data for different solar power plants

| Technology data              |                            | CSP plant | CSP plant with gas-firing | CSP plant with storage | Solar PV with storage, 1h | Solar PV with storage, 2h | Solar PV, roof top | Solar PV plant utility scale |
|------------------------------|----------------------------|-----------|---------------------------|------------------------|---------------------------|---------------------------|--------------------|------------------------------|
| Parameter                    |                            |           |                           |                        |                           |                           |                    |                              |
| Efficiency [%]               |                            | 100       | 100                       | 100                    | 100                       | 100                       | 100                | 100                          |
| Power plant lifetime [years] |                            | 25        | 25                        | 25                     | 20                        | 20                        | 20                 | 25                           |
| Island Countries             | Capacity-/plant factor [%] | 69.10     | 85                        | 92.41                  | 39.72                     | 28.85                     | 43.64              | 54.55                        |
|                              | Day time                   |           |                           |                        |                           |                           |                    |                              |
|                              | Capacity-/plant factor [%] | 13.32     | 85                        | 82.59                  | 17.18                     | 32.29                     | 0                  | 0                            |
|                              | Peak time                  |           |                           |                        |                           |                           |                    |                              |
|                              | Capacity-/plant factor [%] | 0         | 85                        | 0                      | 0                         | 12.71                     | 0                  | 0                            |
|                              | Night time                 |           |                           |                        |                           |                           |                    |                              |
| Inland Countries             | Capacity-/plant factor [%] | 73.69     | 85                        | 73.74                  | 33.19                     | 37.90                     | 35.35              | 52.63                        |
|                              | Part 1                     |           |                           |                        |                           |                           |                    |                              |
|                              | Capacity-/plant factor [%] | 0         | 85                        | 89.45                  | 15.17                     | 49.21                     | 0                  | 0                            |
|                              | Part 2                     |           |                           |                        |                           |                           |                    |                              |
|                              | Capacity-/plant factor [%] | 0         | 85                        | 37.06                  | 0                         | 15.89                     | 0                  | 0                            |
|                              | Part 3                     |           |                           |                        |                           |                           |                    |                              |
| Availability factor [%]      |                            | 100       | 93                        | 100                    | 100                       | 100                       | 100                | 100                          |
| Capital Cost [MUSD/GW]       |                            | 3646.52   | 1687.49                   | 8508.54                | 4258                      | 6275                      | 2100               | 2200                         |
| Variable O&M [MUSD/TWh]      |                            | 6.20      | 4.56                      | 4.56                   | 5.29                      | 4.76                      | 6.61               | 5.58                         |
| Fixed O&M [MUSD/GW]          |                            | 0         | 0                         | 0                      | 0                         | 0                         | 0                  | 0                            |
| Reference:                   |                            | 7         | 7                         | 7                      | 7                         | 7                         | 7                  | 7                            |

## APPENDIX C - COUNTRY SPECIFIC DATA

There are several parameters that variants depending on which country it is, following tables consider the land specific data. References can be found in *Appendix D – References for appendix B & C*.

### C.1 Year split

Table 9: Two different year split has been used, one for the inland countries and one for the island countries.

| Name                                     | Description                             | Share  |
|------------------------------------------|-----------------------------------------|--------|
| <b>Used for Egypt and Zimbabwe</b>       |                                         |        |
| S1D1                                     | Season 1, Day part 1                    | 0.1356 |
| S1D2                                     | Season 1, Day part 2                    | 0.0565 |
| S1D3                                     | Season 1, Day part 3                    | 0.0791 |
| S2D1                                     | Season 2, Day part 1                    | 0.1123 |
| S2D2                                     | Season 2, Day part 2                    | 0.0468 |
| S2D3                                     | Season 2, Day part 3                    | 0.0655 |
| S3D1                                     | Season 3, Day part 1                    | 0.1260 |
| S3D2                                     | Season 3, Day part 2                    | 0.0525 |
| S3D3                                     | Season 3, Day part 3                    | 0.0735 |
| S4D1                                     | Season 4, Day part 1                    | 0.1260 |
| S4D2                                     | Season 4, Day part 2                    | 0.0525 |
| S4D3                                     | Season 4, Day part 3                    | 0.0735 |
| <b>Used for Madagascar and Mauritius</b> |                                         |        |
| SED                                      | Summer Weekend Day time (06:00-17:00)   | 0.0649 |
| SEP                                      | Summer Weekend Peak time (17:00-23:00)  | 0.0354 |
| SEN                                      | Summer Weekend Night time (23:00-06:00) | 0.0413 |
| SWD                                      | Summer Week Day time (06:00-17:00)      | 0.1623 |
| SWP                                      | Summer Week Peak time (17:00-23:00)     | 0.0886 |
| SWN                                      | Summer Week Night time (23:00-06:00)    | 0.1033 |
| WED                                      | Winter Weekend Day time (06:00-17:00)   | 0.0660 |
| WEP                                      | Winter Weekend Peak time (17:00-23:00)  | 0.0360 |
| WEN                                      | Winter Weekend Night time (23:00-06:00) | 0.0420 |
| WWD                                      | Winter Week Day time (06:00-17:00)      | 0.1650 |
| WWP                                      | Winter Week Peak time (17:00-23:00)     | 0.0900 |
| WWN                                      | Winter Week Night time (23:00-06:00)    | 0.1050 |

## C.2 Egypt specific parameters

The possible technologies that are not presented in the table have no residual capacity but they are still included in the model. Please observe that the installed Solar PV capacity is too small therefore will OSeMOSYS programme neglect the power plant.

Table 10: Residual Capacity for different power plants in Egypt

| Residual Capacity [GW] | Diesel DE | HFO OCGT | NG CCGT | NG OCGT | Wind, 30% | Biomass | Solar PV plant utility scale |
|------------------------|-----------|----------|---------|---------|-----------|---------|------------------------------|
| Year                   |           |          |         |         |           |         |                              |
| 2010                   | 0.1445    | 1.6104   | 8.6101  | 7.7662  | 0.4235    | 2.4327  | 0.0001                       |
| 2011                   | 0.1445    | 1.2922   | 8.6101  | 7.4972  | 0.4235    | 2.4327  | 0.00008                      |
| 2012                   | 0.1445    | 1.2777   | 8.6101  | 7.1972  | 0.4235    | 2.4327  | 0.00007                      |
| 2013                   | 0.1445    | 0.9427   | 8.4141  | 7.1972  | 0.4235    | 2.4327  | 0.00006                      |
| 2014                   | 0.0971    | 0.942    | 8.4141  | 6.8972  | 0.4235    | 2.4327  | 0.00006                      |
| 2015                   | 0.0971    | 0.9407   | 8.2796  | 6.8972  | 0.4235    | 2.4327  | 0.00006                      |
| 2016                   | 0.0971    | 0.9252   | 8.2536  | 6.2172  | 0.4235    | 2.4327  | 0.00002                      |
| 2017                   | 0.0485    | 0.6033   | 8.2536  | 6.2172  | 0.4235    | 2.4327  | 0.00002                      |
| 2018                   | 0.0485    | 0.6033   | 8.2536  | 6.0072  | 0.4225    | 2.4327  | 0.00002                      |
| 2019                   | 0.0485    | 0.6033   | 7.1237  | 6.0072  | 0.4225    | 2.4327  | 0                            |
| 2020                   | 0.0485    | 0.6033   | 6.8597  | 5.1372  | 0.4207    | 2.4327  | 0                            |
| 2021                   | 0.0485    | 0.4632   | 6.8597  | 5.1372  | 0.4195    | 2.4327  | 0                            |
| 2022                   | 0.0485    | 0.0835   | 6.8597  | 5.1372  | 0.4195    | 2.4107  | 0                            |
| 2023                   | 0.0243    | 0.0804   | 6.4467  | 4.4678  | 0.4195    | 2.4107  | 0                            |
| 2024                   | 0.0243    | 0.072    | 6.3367  | 3.4978  | 0.4195    | 2.4107  | 0                            |
| 2025                   | 0         | 0.066    | 6.1063  | 2.7378  | 0.4093    | 2.3657  | 0                            |
| 2026                   | 0         | 0        | 6.1063  | 2.3578  | 0.3865    | 2.3657  | 0                            |
| 2027                   | 0         | 0        | 6.1063  | 1.3138  | 0.3535    | 2.3657  | 0                            |
| 2028                   | 0         | 0        | 6.1063  | 0.2218  | 0.2763    | 2.3657  | 0                            |
| 2029                   | 0         | 0        | 6.0543  | 0.2218  | 0.2763    | 2.3657  | 0                            |
| 2030                   | 0         | 0        | 6.0543  | 0.0242  | 0.2763    | 2.3657  | 0                            |
| Reference:             | 7         | 7        | 7       | 7       | 7         | 7       | 7                            |

Table 11: Specific technology data for Egypt power plants

| Technology data              |           | Hydro, Aswan<br>High dam | Hydro, Aswan | Hydro, Esna | Hydro, Faiyun | Hydro, Nagaa Mamadi |
|------------------------------|-----------|--------------------------|--------------|-------------|---------------|---------------------|
| Residual Capacity [GW]       |           |                          |              |             |               |                     |
| Year:                        | 2010-2030 | 2.1                      | 0.592        | 0.08568     | 0.0008        | 0.064               |
| Reference:                   |           | 7                        | 7            | 7           | 7             | 7                   |
| Performance:                 | Year      |                          |              |             |               |                     |
| Efficiency [%]               | 2010      | 100                      | 100          | 100         | 100           | 100                 |
| Power plant lifetime [years] | 2010      | 50                       | 50           | 50          | 50            | 50                  |
| Capacity-/plant factor [%]   | 2010      | 46                       | 46           | 46          | 46            | 46                  |
| Availability factor [%]      | 2010      | 95                       | 95           | 95          | 95            | 95                  |
| Capital Cost [MUSD/GW]       | 2010      | 0                        | 0            | 0           | 0             | 0                   |
| Variable O&M [MUSD/TWh]      | -         | 0.32                     | 0.32         | 0.32        | 0.32          | 0.32                |
| Fixed O&M [MUSD/TWh]         | -         | 21                       | 21           | 21          | 21            | 21                  |
| Reference:                   |           | 9                        | 9            | 9           | 9             | 9                   |

The new installed capacities are presented in the table below, along with corresponding years of installations.

Table 12: Specific technology data for power plants in Egypt

| Technology data              | Diesel | NG CCGT | NG OCGT | Uranium | Wind  | Solar PV Utility scale | HYydro, Gabal Galala | Hydro, Asyut | Hydro, Zefta |
|------------------------------|--------|---------|---------|---------|-------|------------------------|----------------------|--------------|--------------|
| Planned new Capacity [GW]    |        |         |         |         |       |                        |                      |              |              |
| Year                         |        |         |         |         |       |                        |                      |              |              |
| 2014                         | 0.2    | -       | -       | -       | 0.206 | 0.1                    | -                    | -            | -            |
| 2015                         | 2.9    | 4.03    | 1.5     | -       | -     | -                      | -                    | -            | -            |
| 2016                         | 3.25   | -       | -       | -       | -     | -                      | 0.65                 | -            | 0.06         |
| 2017                         | 3.25   | -       | -       | -       | -     | -                      | -                    | 0.032        | -            |
| 2020                         | -      | -       | -       | 1       | -     | -                      | -                    | -            | -            |
| Reference:                   | 7      | 7       | 7       | 7       | 7     | 7                      | 7                    | 7            | 7            |
| Performance:                 |        |         |         |         |       |                        |                      |              |              |
| Efficiency [%]               | -      | -       | -       | -       | -     | -                      | 100                  | 100          | 100          |
| Power plant lifetime [years] | -      | -       | -       | -       | -     | -                      | 50                   | 50           | 50           |
| Capacity-/plant factor [%]   | -      | -       | -       | -       | -     | -                      | 46                   | 46           | 46           |
| Availability factor [%]      | -      | -       | -       | -       | -     | -                      | 95                   | 95           | 95           |
| Capital Cost [MUSD/GW]       | -      | -       | -       | -       | -     | -                      | 0                    | 0            | 0            |
| Variable O&M [MUSD/TWh]      | -      | -       | -       | -       | -     | -                      | 0.32                 | 0.32         | 0.32         |
| Fixed O&M [MUSD/TWh]         | -      | -       | -       | -       | -     | -                      | 21                   | 21           | 21           |
| Reference:                   |        |         |         |         |       |                        | 9                    | 9            | 9            |

Egypt's electricity demand is separated by KTH-dESA in three parts: industrial, rural and urban.

Table 13: Egypt's demand split

| Projected Electricity Demand per Year [PJ] | Industrial Demand | Rural Demand | Rural Demand |
|--------------------------------------------|-------------------|--------------|--------------|
| 2010                                       | 253.4536          | 102.1911     | 102.3999     |
| 2011                                       | 253.4536          | 102.1911     | 102.3999     |
| 2012                                       | 264.1506          | 106.5041     | 106.7216     |
| 2013                                       | 275.1403          | 110.9351     | 111.1619     |
| 2014                                       | 286.5525          | 115.5362     | 115.7724     |
| 2015                                       | 298.5441          | 120.3713     | 120.6173     |
| 2016                                       | 311.2137          | 125.4795     | 125.7359     |
| 2017                                       | 329.7865          | 132.9681     | 133.2399     |
| 2018                                       | 350.4214          | 141.2882     | 141.5768     |
| 2019                                       | 370.7939          | 149.5021     | 149.8077     |
| 2020                                       | 392.1116          | 158.0972     | 158.4204     |
| 2021                                       | 414.8276          | 167.2562     | 167.5981     |
| 2022                                       | 438.6308          | 176.8536     | 177.2151     |
| 2023                                       | 463.6613          | 186.9457     | 187.3276     |
| 2024                                       | 490.0975          | 197.6046     | 198.0086     |
| 2025                                       | 519.5074          | 205.4624     | 209.8907     |
| 2026                                       | 550.2694          | 221.8656     | 222.319      |
| 2027                                       | 582.7936          | 234.9792     | 235.4594     |
| 2028                                       | 617.1311          | 248.8241     | 249.3324     |
| 2029                                       | 653.3381          | 263.4224     | 263.9607     |
| 2030                                       | 691.5229          | 278.8183     | 279.3882     |
| Reference:                                 | 7                 | 7            | 7            |

Egypt has grid interconnections with Libya and Jordan it also plans to build interconnections with Sudan and Saudi Arabia, parameters used in the simulations are tabulated below.

**Table 14: Technology data for Egypt's interconnections with other countries for export/import of electricity**

| Technology data                  |      | Export/Import<br>Egypt-Sudan | Export/Import<br>Egypt-Libya | Export/Import<br>Egypt-Jordan | Export/Import<br>Egypt-Saudi Arabia |
|----------------------------------|------|------------------------------|------------------------------|-------------------------------|-------------------------------------|
| Parameter                        | Year |                              |                              |                               |                                     |
| Residual Capacity [GW]           |      |                              |                              |                               |                                     |
|                                  | 2010 | 0                            | 0.24                         | 0.6                           | 0                                   |
| New Capacity [GW]                |      |                              |                              |                               |                                     |
|                                  | 2015 | -                            | -                            | -                             | 3                                   |
|                                  | 2016 | 2                            | -                            | -                             | -                                   |
| Performance                      |      |                              |                              |                               |                                     |
| Efficiency [%]                   | 2010 | 100                          | 100                          | 100                           | 100                                 |
| Power plant lifetime [years]     | 2010 | 60                           | 60                           | 60                            | 60                                  |
| Capacity factor/plant factor [%] | 2010 | 100                          | 100                          | 100                           | 100                                 |
| Availability factor [%]          | 2010 | 99                           | 99                           | 99                            | 99                                  |
| Capital Cost [MUSD/GW]           | 2010 | 860.81                       | -                            | -                             | 516.95                              |
| Variable O&M [MUSD/TWh]          | -    | 22.22                        | 22.22                        | 22.22                         | 22.22                               |
| Fixed O&M [MUSD/GW]              | -    | 0                            | 0                            | 0                             | 0                                   |
| Reference:                       |      | 10                           | 10                           | 10                            | 10                                  |

**Table 15: Specific efficiency for distribution and transmission in Egypt**

| Data for all technologies |      | Distribution to<br>Industry | Distribution to<br>Rural Areas | Distribution to<br>Urban Areas | Transmission |
|---------------------------|------|-----------------------------|--------------------------------|--------------------------------|--------------|
| Parameter                 | Year |                             |                                |                                |              |
| Efficiency [%]            | 2010 | 95                          | 70                             | 95                             | 96.30        |
| Reference:                |      | 3                           | 3                              | 3                              | 3            |

### C.3 Madagascar specific parameters

The possible technologies that are not presented in the table have no residual capacity but they are still included in the model.

Table 16: Madagascar's residual capacity for fossil fuels

| Residual Capacity [GW] | Diesel DE | HFO OCGT |
|------------------------|-----------|----------|
| Year                   |           |          |
| 2010                   | 0.2611    | 0.1181   |
| 2011                   | 0.2611    | 0.1181   |
| 2012                   | 0.2334    | 0.0886   |
| 2013                   | 0.2334    | 0.0886   |
| 2014                   | 0.2334    | 0.0886   |
| 2015                   | 0.2334    | 0.0886   |
| 2016                   | 0.2334    | 0.0886   |
| 2017                   | 0.2334    | 0.0886   |
| 2018                   | 0.2334    | 0.0886   |
| 2019                   | 0.2334    | 0.0886   |
| 2020                   | 0.2214    | 0.0886   |
| 2021                   | 0.2164    | 0.0886   |
| 2022                   | 0.2164    | 0.0886   |
| 2023                   | 0.2163    | 0.0886   |
| 2024                   | 0.1963    | 0.0886   |
| 2025                   | 0.1963    | 0.0886   |
| 2026                   | 0.1959    | 0.0886   |
| 2027                   | 0.1909    | 0.0886   |
| 2028                   | 0.1719    | 0.0886   |
| 2029                   | 0.1719    | 0.0886   |
| 2030                   | 0.1469    | 0.0886   |
| Reference:             | 7         | 7        |

Table 17: Land specific data for hydro power plants in Madagascar

| Technology data               | Hydro, mini | Hydro, Andekaleka | Hydro, Antelomita | Hydro, Manandrona | Hydro, Mandraka | Hydro, Maroantsetra | Hydro, Namorona | Hydro, Sahanivotry | Hydro, Volobe |
|-------------------------------|-------------|-------------------|-------------------|-------------------|-----------------|---------------------|-----------------|--------------------|---------------|
| Total installed Capacity [GW] |             |                   |                   |                   |                 |                     |                 |                    |               |
| Year                          |             |                   |                   |                   |                 |                     |                 |                    |               |
| 2010-2030                     | 0.0023      | 0.0014            | 0.0097            | 0.0014            | 0.0240          | 0.0024              | 0.0050          | 0.0150             | 0.0071        |
| Reference:                    | 7           | 7                 | 7                 | 7                 | 7               | 7                   | 7               | 7                  | 7             |

The new installed capacities are presented in the table below.

Table 18: the new capacities in Madagascar

| Technology data           | Coal | Hydro, AMBODIROKA | Hydro, SAHOIFIKA | Hydro, TSIAZOMPANIRY FRAISE |
|---------------------------|------|-------------------|------------------|-----------------------------|
| Planned new capacity [GW] |      |                   |                  |                             |
| Year                      |      |                   |                  |                             |
| 2011                      | 0.12 | -                 | -                | -                           |
| 2012                      | -    | -                 | -                | 0.0056                      |
| 2013                      | -    | 0.0405            | -                | -                           |
| 2015                      | -    | -                 | 0.1050           | -                           |
| Reference:                | 7    | 7                 | 7                | 7                           |

Table 19: Demand split in Madagascar

| Projected Electricity Demand per Year [PJ] | Industrial Demand | Other consumers Demand |
|--------------------------------------------|-------------------|------------------------|
| 2010                                       | 1.26              | 4.0532                 |
| 2011                                       | 1.3532            | 4.3531                 |
| 2012                                       | 1.4574            | 4.6883                 |
| 2013                                       | 1.558             | 5.0118                 |
| 2014                                       | 1.6749            | 5.3877                 |
| 2015                                       | 1.7904            | 5.7594                 |
| 2016                                       | 1.9157            | 6.1626                 |
| 2017                                       | 2.0498            | 6.594                  |
| 2018                                       | 2.1913            | 7.0489                 |
| 2019                                       | 2.3469            | 7.5494                 |
| 2020                                       | 2.5112            | 8.0779                 |
| 2021                                       | 2.6869            | 8.6433                 |
| 2022                                       | 2.875             | 9.2484                 |
| 2023                                       | 3.0763            | 9.8957                 |
| 2024                                       | 3.2916            | 10.5884                |
| 2025                                       | 3.522             | 11.3296                |
| 2026                                       | 3.7686            | 12.1227                |
| 2027                                       | 4.0324            | 12.9713                |
| 2028                                       | 4.3146            | 13.8793                |
| 2029                                       | 4.6166            | 14.8508                |
| 2030                                       | 4.9398            | 15.8904                |
| Reference:                                 | 5                 | 5                      |

Table 20: Specific efficiency for distribution and transmission in Madagascar

| Technology data |      | Distribution to Industry | Distribution to Other consumers | Transmission |
|-----------------|------|--------------------------|---------------------------------|--------------|
| Parameter       | Year |                          |                                 |              |
| Efficiency [%]  | 2010 | 93.39                    | 93.39                           | 93           |
| Reference:      |      | 3                        | 3                               | 3            |

## C.4 Mauritius specific parameters

Table 21: The residual capacity in Mauritius

| Residual Capacity [GW] | Bagasse | Bagasse/coal | Diesel | HFO    | Wind, 30% | Solar PV plant utility scale |
|------------------------|---------|--------------|--------|--------|-----------|------------------------------|
| Year                   |         |              |        |        |           |                              |
| 2010                   | 0.2257  | 0.1985       | 0.0873 | 0.4768 | 0.0002    | 0.007                        |
| 2011                   | 0.1389  | 0.1985       | 0.0863 | 0.2598 | 0.0002    | 0.007                        |
| 2012                   | 0.1389  | 0.1985       | 0.0858 | 0.2598 | 0.0002    | 0.007                        |
| 2013                   | 0.1389  | 0.1985       | 0.0658 | 0.2598 | 0.0002    | 0.007                        |
| 2014                   | 0.1389  | 0.1768       | 0.0653 | 0.2164 | 0.0002    | 0.007                        |
| 2015                   | 0.1389  | 0.1768       | 0.0653 | 0.2164 | 0.0002    | 0.007                        |
| 2016                   | 0.1389  | 0.1768       | 0.0385 | 0.2164 | 0.0002    | 0.007                        |
| 2017                   | 0.1389  | 0.1768       | 0.0385 | 0.1947 | 0.0002    | 0.007                        |
| 2018                   | 0.1389  | 0.1768       | 0.038  | 0.1707 | 0.0002    | 0.007                        |
| 2019                   | 0.1389  | 0.1768       | 0.038  | 0.1707 | 0.0002    | 0.007                        |
| 2020                   | 0.1172  | 0.1768       | 0      | 0.1397 | 0.0002    | 0.007                        |
| 2021                   | 0.1172  | 0.1768       | 0      | 0.1382 | 0.0002    | 0.007                        |
| 2022                   | 0.1172  | 0.1768       | 0      | 0.1372 | 0.0002    | 0.007                        |
| 2023                   | 0.1172  | 0.1768       | 0      | 0.1362 | 0.0002    | 0.007                        |
| 2024                   | 0.1172  | 0.1768       | 0      | 0.1057 | 0.0002    | 0.007                        |
| 2025                   | 0.1172  | 0.1768       | 0      | 0.0752 | 0.0002    | 0                            |
| 2026                   | 0.1172  | 0.1768       | 0      | 0.0752 | 0.0002    | 0                            |
| 2027                   | 0.0738  | 0.1768       | 0      | 0.0752 | 0.0002    | 0                            |
| 2028                   | 0.002   | 0.1551       | 0      | 0.0752 | 0         | 0                            |
| 2029                   | 0.002   | 0.1551       | 0      | 0.0714 | 0         | 0                            |
| 2030                   | 0.002   | 0.1117       | 0      | 0.0714 | 0         | 0                            |
| Reference:             | 7       | 7            | 7      | 7      | 7         | 7                            |

Table 22: The residual capacity for hydro power in Mauritius

| Technology data        | Hydro, Tamarin Falls | Hydro, La Ferme | Hydro, Magenta Tamarin | Hydro, Le Va (Eau Bleue) | Hydro, Cascade Cecile | Hydro, Ferney | Hydro, Champagne Maudslayi | Hydro, Reduit Rebuild |
|------------------------|----------------------|-----------------|------------------------|--------------------------|-----------------------|---------------|----------------------------|-----------------------|
| Residual Capacity [GW] |                      |                 |                        |                          |                       |               |                            |                       |
| Year                   |                      |                 |                        |                          |                       |               |                            |                       |
| 2010-2030              | 0.0695               | 0.0217          | 0.0217                 | 0.0434                   | 0.0217                | 0.0434        | 0.0367                     | 0.0012                |
| Reference:             | 7                    | 7               | 7                      | 7                        | 7                     | 7             | 7                          | 7                     |

Table 23: New planned hydro power in Mauritius

| Technology data           | Hydro, Bois Cherie | Hydro, Highlands Estate | Hydro, Riche-En-Eau | Hydro, St-Felix |
|---------------------------|--------------------|-------------------------|---------------------|-----------------|
| Planned new capacity [GW] |                    |                         |                     |                 |
| Year                      |                    |                         |                     |                 |
| 2010-2030                 | 0.0001             | 0.0004                  | 0.0001              | 0.0001          |
| Reference:                | 7                  | 7                       | 7                   | 7               |

Table 24: New capacity in Mauritius

| Technology data           | Biogas | Coal | HFO    | Wind   | Solar PV Utility scale |
|---------------------------|--------|------|--------|--------|------------------------|
| Planned new Capacity [GW] |        |      |        |        |                        |
| Year                      |        |      |        |        |                        |
| 2012                      | 0.3    | -    | 0.0625 | -      | -                      |
| 2013                      | -      | -    | -      | 0.0280 | -                      |
| 2015                      | -      | 0.5  | -      | 0.0384 | -                      |
| 2016                      | -      | 0.5  | -      | 0.0100 | -                      |
| 2017                      | -      | -    | -      | -      | 0.2                    |
| 2019                      | -      | -    | -      | 0.0100 | -                      |
| 2020                      | -      | -    | -      | -      | 0.2                    |
| 2022                      | -      | -    | -      | 0.0100 | -                      |
| 2023                      | -      | -    | -      | -      | 0.2                    |
| 2025                      | -      | -    | -      | 0.0100 | -                      |
| 2026                      | -      | -    | -      | -      | 0.2                    |
| 2028                      | -      | -    | -      | 0.0100 | -                      |
| 2029                      | -      | -    | -      | 0.0200 | 0.2                    |
| Reference:                | 7      | 7    | 7      | 7      | 7                      |

Table 25: Mauritius demand spilt

| Projected Electricity Demand per Year [PJ] | Industrial Demand | Other consumers Demand |
|--------------------------------------------|-------------------|------------------------|
| 2010                                       | 3.3624            | 5.3604                 |
| 2011                                       | 3.4569            | 5.5642                 |
| 2012                                       | 3.554             | 5.7756                 |
| 2013                                       | 3.6539            | 5.9948                 |
| 2014                                       | 3.7566            | 6.2221                 |
| 2015                                       | 3.8621            | 6.4578                 |
| 2016                                       | 3.9706            | 6.7023                 |
| 2017                                       | 4.0822            | 6.9557                 |
| 2018                                       | 4.1969            | 7.2185                 |
| 2019                                       | 4.3149            | 7.491                  |
| 2020                                       | 4.4361            | 7.7735                 |
| 2021                                       | 4.5608            | 8.0664                 |
| 2022                                       | 4.6889            | 8.3701                 |
| 2023                                       | 4.8207            | 8.6849                 |
| 2024                                       | 4.9561            | 9.0114                 |
| 2025                                       | 5.0954            | 9.3498                 |
| 2026                                       | 5.2386            | 9.7006                 |
| 2027                                       | 5.3858            | 10.0644                |
| 2028                                       | 5.5371            | 10.4414                |
| 2029                                       | 5.6927            | 10.8323                |
| 2030                                       | 5.8527            | 11.2375                |
| Reference:                                 | 5                 | 5                      |

Table 26: Specific efficiency for distribution and transmission in Mauritius

| Technology data |      | Distribution to Industry | Distribution to Other consumers | Transmission |
|-----------------|------|--------------------------|---------------------------------|--------------|
| Parameter       | Year |                          |                                 |              |
| Efficiency [%]  | 2010 | 92.91                    | 92.91                           | 93           |
| Reference:      |      | 3                        | 3                               | 3            |

## C.5 Zimbabwe specific parameters

Table 27: Zimbabwe's residual fossil capacity

| Residual Capacity [GW] | Diesel DE | Coal   | Biomass |
|------------------------|-----------|--------|---------|
| Year                   |           |        |         |
| 2010                   | 0.0143    | 1.1175 | 0.0815  |
| 2011                   | 0.0143    | 1.1175 | 0.0815  |
| 2012                   | 0.0143    | 1.1175 | 0.0615  |
| 2013                   | 0.0143    | 1.1175 | 0.0615  |
| 2014                   | 0.0143    | 0.6800 | 0.0615  |
| 2015                   | 0.0143    | 0.5600 | 0.0615  |
| 2016                   | 0.0143    | 0.2200 | 0.0615  |
| 2017                   | 0.0143    | 0      | 0.0615  |
| 2018                   | 0.0143    | 0      | 0.0615  |
| 2019                   | 0.0143    | 0      | 0.0615  |
| 2020                   | 0.0143    | 0      | 0.0615  |
| 2021                   | 0.0143    | 0      | 0.0615  |
| 2022                   | 0.0143    | 0      | 0.0615  |
| 2023                   | 0.0143    | 0      | 0.0615  |
| 2024                   | 0.0143    | 0      | 0.0615  |
| 2025                   | 0.0143    | 0      | 0.0615  |
| 2026                   | 0.0143    | 0      | 0.0615  |
| 2027                   | 0.0143    | 0      | 0.0415  |
| 2028                   | 0.0143    | 0      | 0.0415  |
| 2029                   | 0.0143    | 0      | 0.0415  |
| 2030                   | 0.0143    | 0      | 0.0415  |
| Reference:             | 7         | 7      | 7       |

Table 28: Residual capacity for Zimbabwe's hydro power plants

| Technology data        | Hydro, ABERFOYLE ESTATES | Hydro, CLAREMONT HYDRO | Hydro, KARIBA SOUTH | Hydro, RUSITU | Hydro, mini (existing capacity) | Hydro, MUTIRIKWI DAM | Hydro, TOKWE-MURKOSI |
|------------------------|--------------------------|------------------------|---------------------|---------------|---------------------------------|----------------------|----------------------|
| Residual Capacity [GW] |                          |                        |                     |               |                                 |                      |                      |
| Year                   |                          |                        |                     |               |                                 |                      |                      |
| 2010-2030              | 0.0004                   | 0.0003                 | 0.6800              | 0.0008        | 0.0001                          | 0.0051               | 0.0050               |
| Reference:             | 7                        | 7                      | 7                   | 7             | 7                               | 7                    | 7                    |

Table 29: Technology data for Zimbabwe's interconnections for import/export of electricity

| Technology data                  |      | Export/Import<br>Zimbabwe |
|----------------------------------|------|---------------------------|
| Parameter                        | Year |                           |
| Residual Capacity [GW]           |      |                           |
|                                  | 2010 | 0.3*                      |
| Performance                      |      |                           |
| Efficiency [%]                   | 2010 | 100                       |
| Power plant lifetime [years]     | 2010 | 60                        |
| Capacity factor/plant factor [%] | 2010 | 100                       |
| Availability factor [%]          | 2010 | 100                       |
| Capital Cost [MUSD/GW]           | 2010 | -                         |
| Variable O&M<br>[MUSD/TWh]       | -    | 2.77778**                 |
| Fixed O&M [MUSD/GW]              | -    | -                         |
| Reference:                       |      | *11<br>**10               |

Table 30: New capacities planned in Zimbabwe

| Technology data           | Coal | Hydro, BATOKA | Hydro, KARIBA SOUTH | Hydro, MUTIRIKWI DAM | Hydro, TOKWE-MURKOSI |
|---------------------------|------|---------------|---------------------|----------------------|----------------------|
| Planned new Capacity [GW] |      |               |                     |                      |                      |
| Year                      |      |               |                     |                      |                      |
| 2013                      | -    | -             | -                   | -                    | 0.005                |
| 2014                      | -    | -             | -                   | 0.005                | -                    |
| 2015                      | 1.2  | 0.3           | 0.3                 | -                    | -                    |
| 2017                      | 0.6  | -             | -                   | -                    | -                    |
| 2018                      | 0.6  | -             | -                   | -                    | -                    |
| 2020                      | 0.6  | -             | -                   | -                    | -                    |
| Reference:                | 7    | 7             | 7                   | 7                    | 7                    |

Table 31: Zimbabwe's demand split

| Projected Electricity Demand per Year [PJ] | Industrial Demand | Other Consumers Demand |
|--------------------------------------------|-------------------|------------------------|
| 2010                                       | 19.7316           | 25.511                 |
| 2011                                       | 20.931            | 27.0616                |
| 2012                                       | 22.2032           | 28.7065                |
| 2013                                       | 23.5528           | 30.4514                |
| 2014                                       | 24.9844           | 32.3023                |
| 2015                                       | 26.5031           | 34.2658                |
| 2016                                       | 28.114            | 36.3486                |
| 2017                                       | 29.8229           | 38.558                 |
| 2018                                       | 31.6356           | 40.9017                |
| 2019                                       | 33.5586           | 43.3878                |
| 2020                                       | 35.5984           | 46.0251                |
| 2021                                       | 37.7622           | 48.8227                |
| 2022                                       | 40.0575           | 51.7903                |
| 2023                                       | 42.4923           | 54.9383                |
| 2024                                       | 45.0752           | 58.2777                |
| 2025                                       | 47.815            | 61.82                  |
| 2026                                       | 50.7214           | 65.5776                |
| 2027                                       | 53.8044           | 69.5637                |
| 2028                                       | 57.0748           | 73.792                 |
| 2029                                       | 60.544            | 78.2773                |
| 2030                                       | 64.2241           | 83.0353                |
| Reference:                                 | 5                 | 5                      |

Table 32: Specific efficiency for distribution and transmission in Zimbabwe

| Technology data |      | Distribution to Industry | Distribution to Others | Transmission |
|-----------------|------|--------------------------|------------------------|--------------|
| Parameter       | Year |                          |                        |              |
| Efficiency [%]  | 2010 | 96.03                    | 96.03                  | 95.99        |
| Reference:      |      | 3                        | 3                      | 3            |

# APPENDIX D - REFERENCE LIST FOR APPENDIX B&C

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| Nr | Reference                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Assumption                                                                                                                                                                                                                                         |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | KTH-dESA (2014a), <i>Model data from Egypt/EAPP model</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Import price is 10 % higher than national resources                                                                                                                                                                                                |
| 2  | EIA-ETSAP (2010), <i>Biomass for Heat and Power</i> , available from: <a href="http://iea-etsap.org/web/E-TechDS/PDF/E05-BiomassforHP-GS-AD-gct.pdf">http://iea-etsap.org/web/E-TechDS/PDF/E05-BiomassforHP-GS-AD-gct.pdf</a><br>[2014-07-02]                                                                                                                                                                                                                                                                                                         | Data for Anaerobic digestion is used.                                                                                                                                                                                                              |
| 3  | International energy statistics, 2014 (2014)<br><a href="http://knoema.com/EIAIES2014/international-energy-statistics-2014?tsId=1231510">http://knoema.com/EIAIES2014/international-energy-statistics-2014?tsId=1231510</a><br>[2014-07-20]                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                    |
| 4  | IEA-ETSAP (2010), <i>Nuclear Power</i> , available at: <a href="http://iea-etsap.org/web/E-TechDS/PDF/E03-Nuclear-Power-GS-AD-gct.pdf">http://iea-etsap.org/web/E-TechDS/PDF/E03-Nuclear-Power-GS-AD-gct.pdf</a><br>[2014-07-02]                                                                                                                                                                                                                                                                                                                      | Mean value, assumed to be for import price, assumed to be 10 % higher than domestic extraction price.                                                                                                                                              |
| 5  | African Regional Energy Statistics (2013), Data for Mauritius, available from: <a href="http://opendataforafrica.org/rpwtdv/african-regional-energy-statistics-2013">http://opendataforafrica.org/rpwtdv/african-regional-energy-statistics-2013</a><br><br>CEB (2012), Demand forecast for Mauritius, Table 4.4, available at: <a href="http://ceb.intnet.mu/CorporateInfo/IEP2013/Chapter4_Demand%20Forecast%20for%20Mauritius.pdf">http://ceb.intnet.mu/CorporateInfo/IEP2013/Chapter4_Demand%20Forecast%20for%20Mauritius.pdf</a><br>[2014-07-02] | Total and industrial energy consumption is available for 2010 as African Regional Statistics. CEB provides forecasts of the compounded annual growth rate for total and industrial electricity demand 2011-2022- Those numbers are used 2011-2030. |
| 6  | KTH-dESA (2014a), <i>Model data from Egypt/EAPP model</i><br><br>CEB (2009), <i>Annual Report 2009</i> , available from: <a href="http://ceb.intnet.mu/">http://ceb.intnet.mu/</a><br>[2014-07-02]                                                                                                                                                                                                                                                                                                                                                    | Data about costs are from KTH-dESA. Losses are based on 2009 data from CEB.                                                                                                                                                                        |
| 7  | KTH-dESA (2014b), Data table "LCOE 2013 03 19_VS_V2-1"<br><br>KTH-dESA (2014a), <i>Model data from Egypt/EAPP model</i><br><br>WEPP (2013), <i>U.D.I PLATTS, World Electric Power Plants Database (WEPP)</i> , Washington DC.<br><br>IEA ETSAP Technology Brief, May 2010<br><br><a href="http://www.iea-etsap.org/web/e-techds/pdf/e07-hydropower-gs-gct.pdf">http://www.iea-etsap.org/web/e-techds/pdf/e07-hydropower-gs-gct.pdf</a>                                                                                                                | Availability factors are from KTH-dESA (2014a), the remaining data from KTH-dESA (2014b). Capacities from WEPP.<br><br>Capital cost and fixed cost for small, medium and large hydro power are from IEA ETSAP.                                     |
| 8  | IEA-ETSAP (2013), <i>Biomass Co-firing</i> , available from: <a href="http://iea-etsap.org/web/E-TechDS/PDF/E21IR_Bio-cofiring_PL_Jan2013_final_GSOK.pdf">http://iea-etsap.org/web/E-TechDS/PDF/E21IR_Bio-cofiring_PL_Jan2013_final_GSOK.pdf</a><br>[2014-07-20]                                                                                                                                                                                                                                                                                      | Data for coal PPs with biomass co-firing from EIA-ETSAP (2013). When ranges are given, typical or mean values                                                                                                                                      |

IEA-ETSAP (2010), *Biomass for Heat and Power*, available from:  
<http://iea-etsap.org/web/E-TechDS/PDF/E05-BiomassforHP-GS-AD-gct.pdf>  
 [2014-07-02]

WEPP (2013), *U.D.I PLATTS, World Electric Power Plants Database (WEPP)*, Washington DC.

Ministry of Renewable Energy & Public Utilities (2009), *DRAFT Long Term Energy Strategy*, available from:  
<http://www.mediaterre.org/docactu,Q0VEUkVGSS9kb2NzL0FMV EVFU3RyYXQtZlJhZnQ=,7.pdf>  
 [2014-02-07]

CEB (2009), *Annual Report 2009*, available from:  
<http://ceb.intnet.mu/>  
 [2014-07-02]

- 9 WEPP (2013), *U.D.I PLATTS, World Electric Power Plants Database (WEPP)*, Washington DC.

IEA-ETSAP (2010), *Hydropower*, available from:  
[http://iea-etsap.org/web/E-TechDS/PDF/E06-hydropower-GS-gct\\_ADfina\\_gs.pdf](http://iea-etsap.org/web/E-TechDS/PDF/E06-hydropower-GS-gct_ADfina_gs.pdf)  
 [2014-07-02]

- 10 ICA (2011), *Regional power status in African power pools*, available from:  
[http://www.icafrica.org/fileadmin/documents/Knowledge/Energy/ICA\\_RegionalPowerPools\\_Report.pdf](http://www.icafrica.org/fileadmin/documents/Knowledge/Energy/ICA_RegionalPowerPools_Report.pdf)  
 [2014-07-20]

- 11 EIA (2011), *International Energy Statistics*, available from:  
<http://www.eia.gov/cfapps/ipdbproject/iedindex3.cfm?tid=2&pid=2&aid=9&cid=EG,&syid=2008&eyid=2012&unit=BKWH>  
 [2014-08-04]

are used. For construction time and availability factor data for Biomass CHP (IEA-ETSAP, 2010) is used.

Capacities from WEPP. The new waste/biomass PP and biogas PP are assumed to be built 2013 (Ministry of E&PU, 2009), capacities from CEB.

Capacities from WEPP. Performance data from ETSAP, considering the size (>10 MW, 1-10 MW and <1 MW) of the hydro plants.

Cost for import and export electricity between countries in the African power pools.

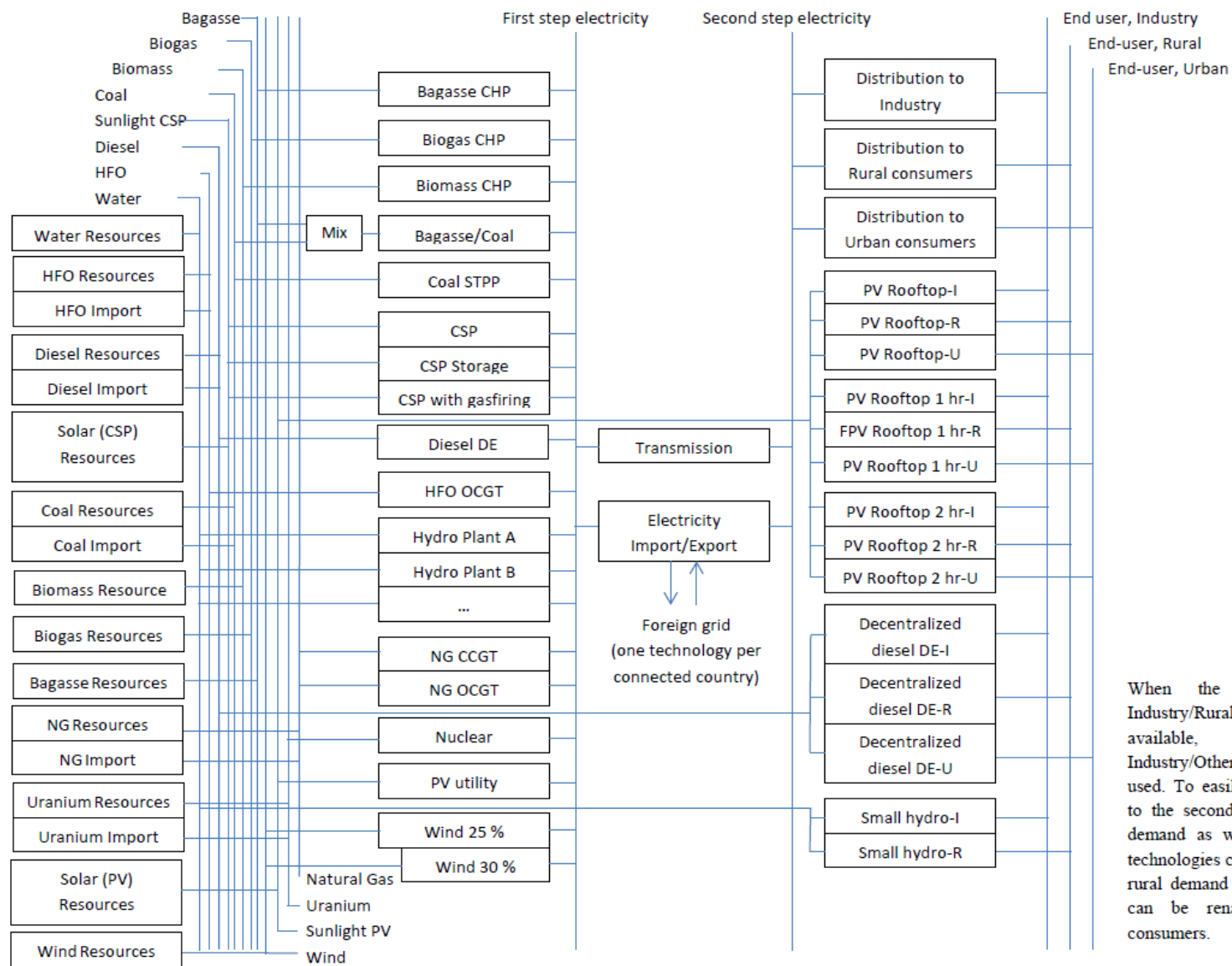
Converted to 2010 USD by CPI Inflation Calculator, available from:  
<http://www.dol.gov/dol/topic/statistics/inflation.htm>  
 [2014-07-01 – 2014-07-15]

When data found is not presented as one single value, the “typical” value is used when available, otherwise a mean value is used.

## **APPENDIX E – DETAILED REFERENCE ENERGY SYSTEM**

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On the following page, a detailed RES is presented. All models are based on the RES, even though all technologies/fuels are not represented in all models.



When the demand split Industry/Rural/Urban is not available, the split Industry/Other consumers is used. To easily adapt the RES to the second split, the urban demand as well as all urban technologies can be deleted and rural demand and technologies can be renamed to Other consumers.

Figure 29: Detailed RES